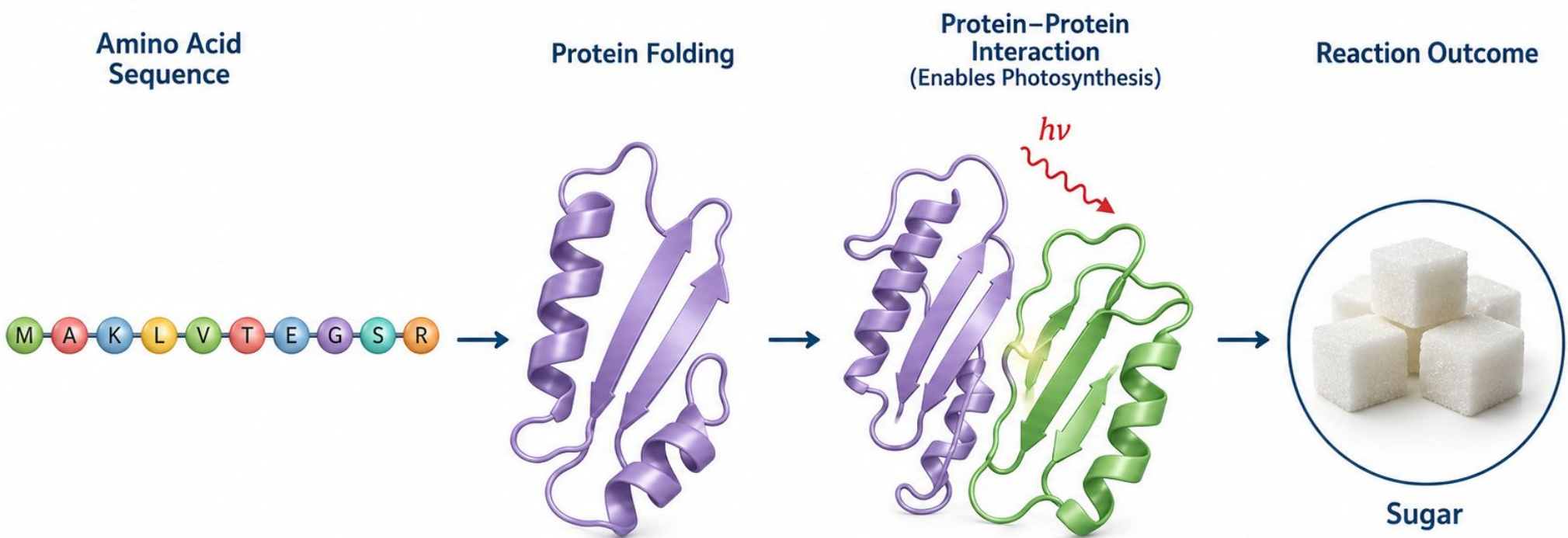


Fine-tuning a Protein Language Model: *Pin the Protein on the Cell*

Background



MAKLVTGSR

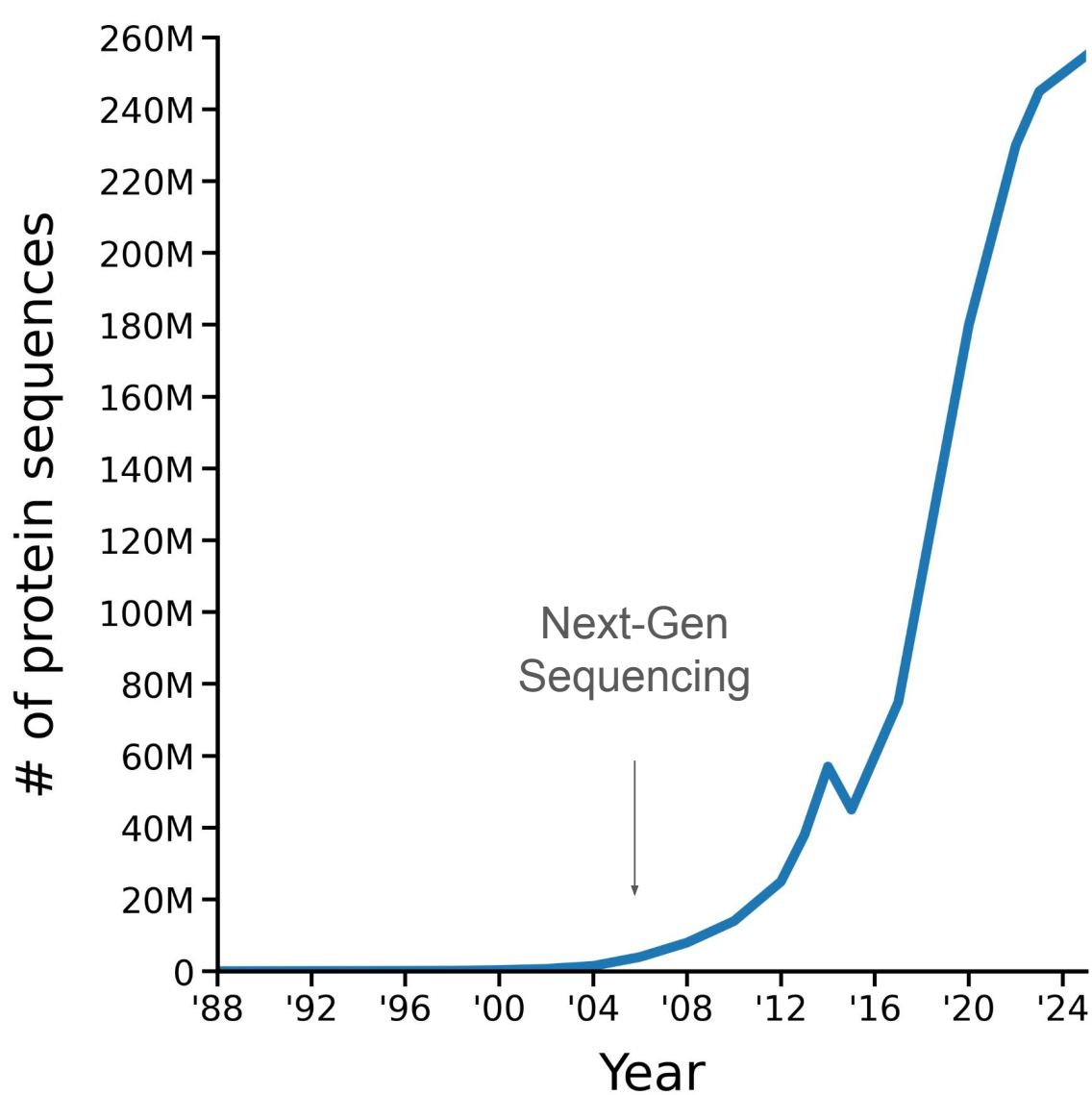
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MAKLVTGSR,LSQWGKLEEQ

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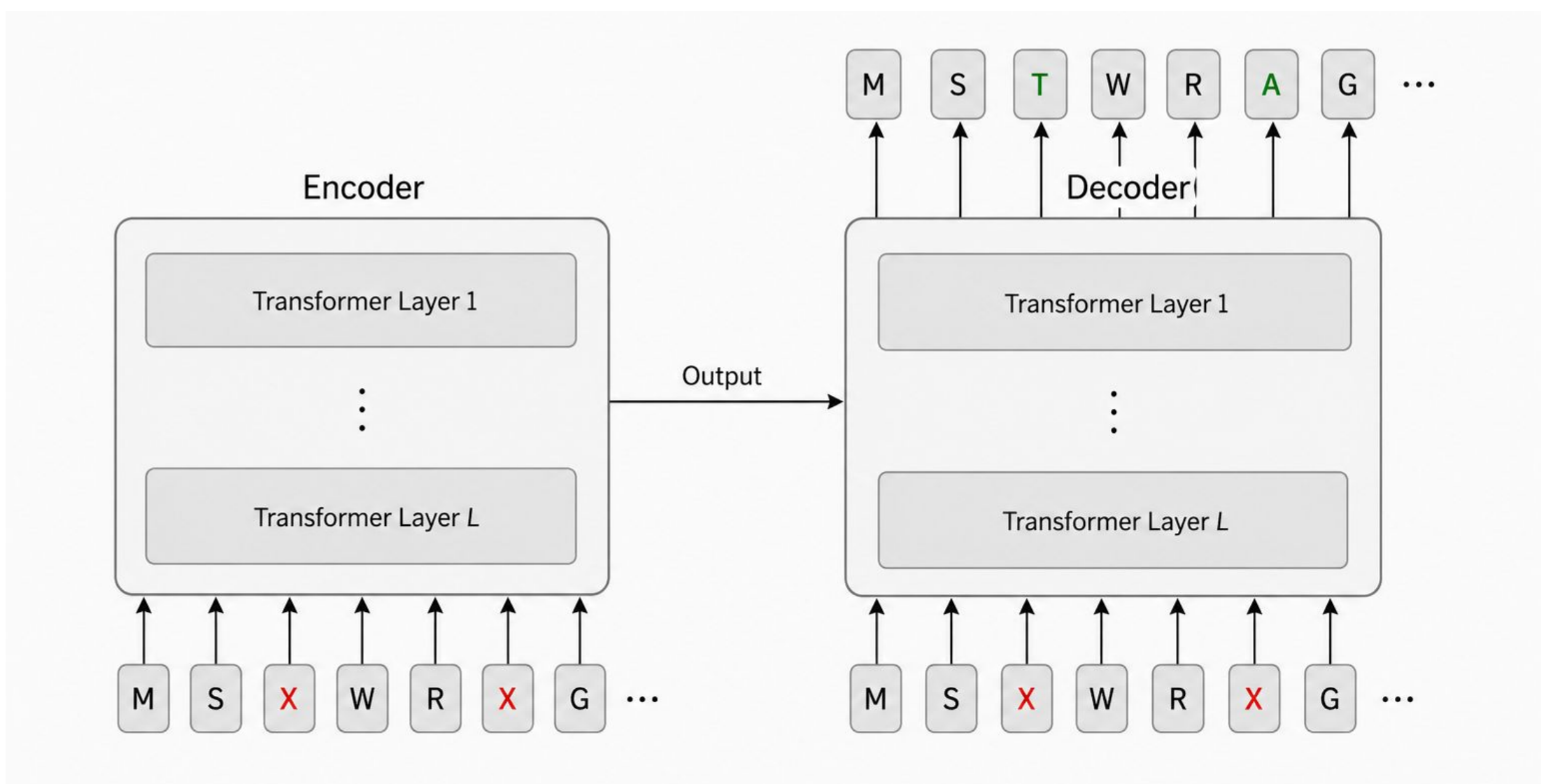
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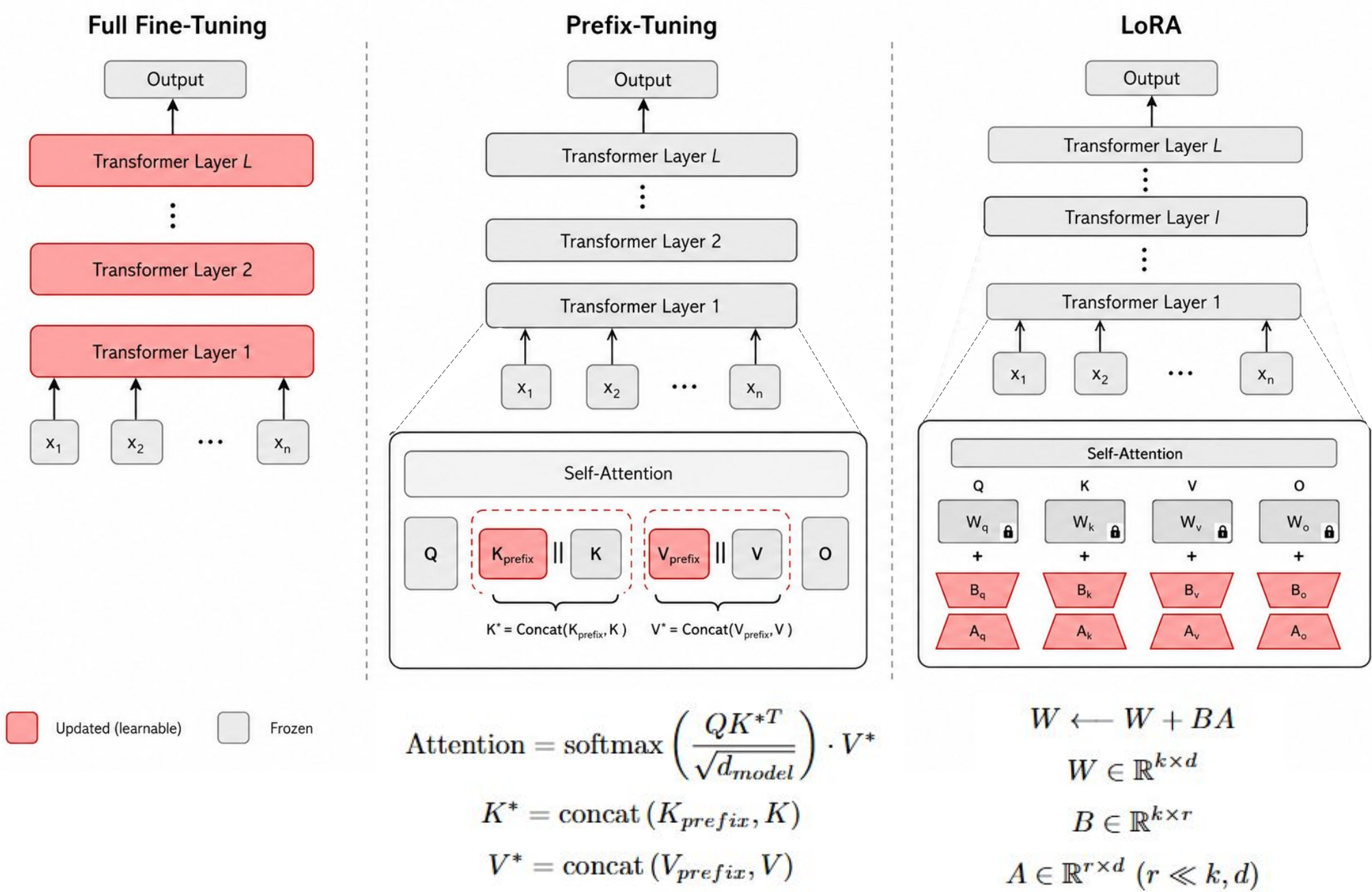
Solution:

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Architecture



ProtT5 architecture and MLM pretraining objective. X = mask token.

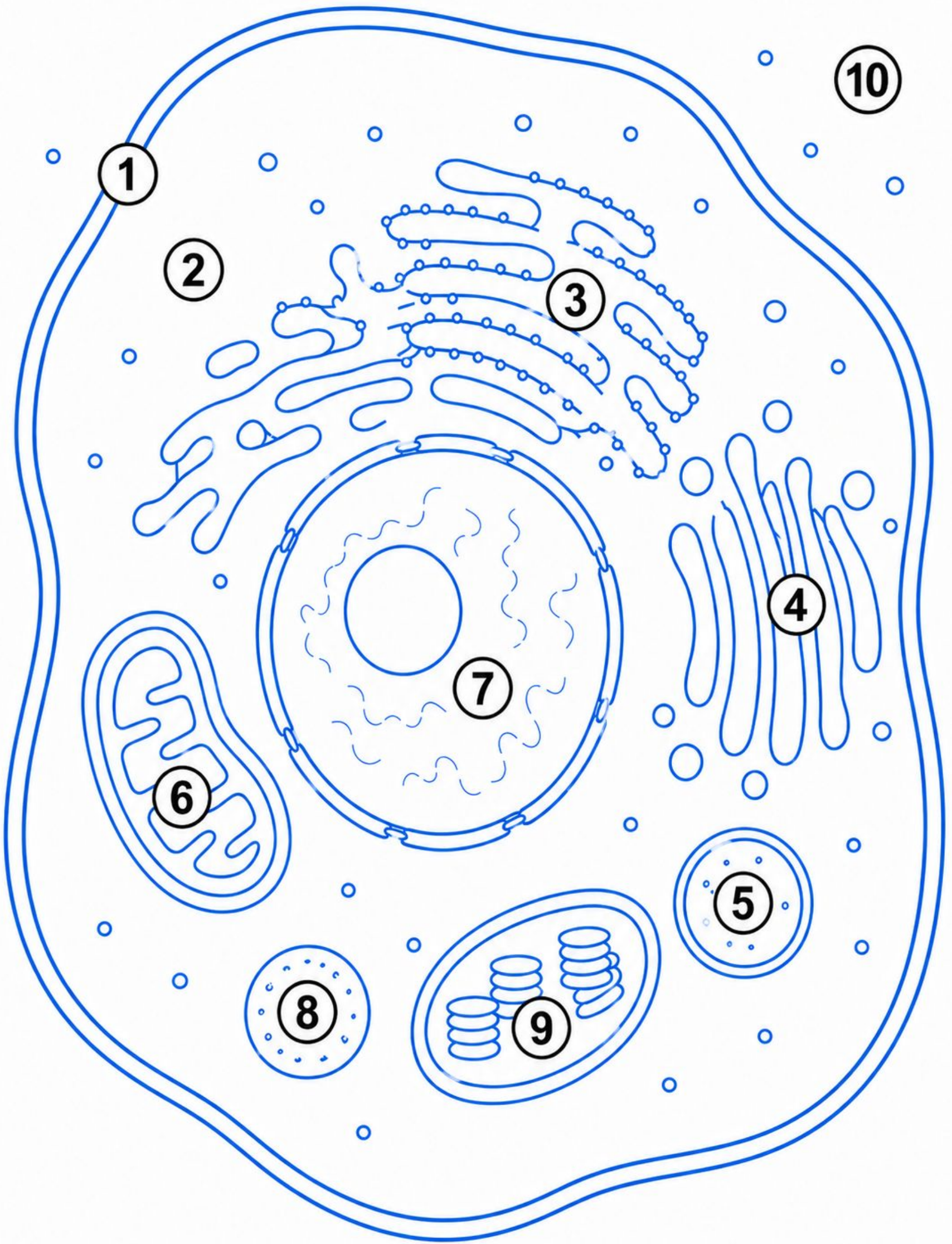


Schematic of Full Fine-Tuning, Prefix-Tuning and Low-Rank Adaptation (LoRA).

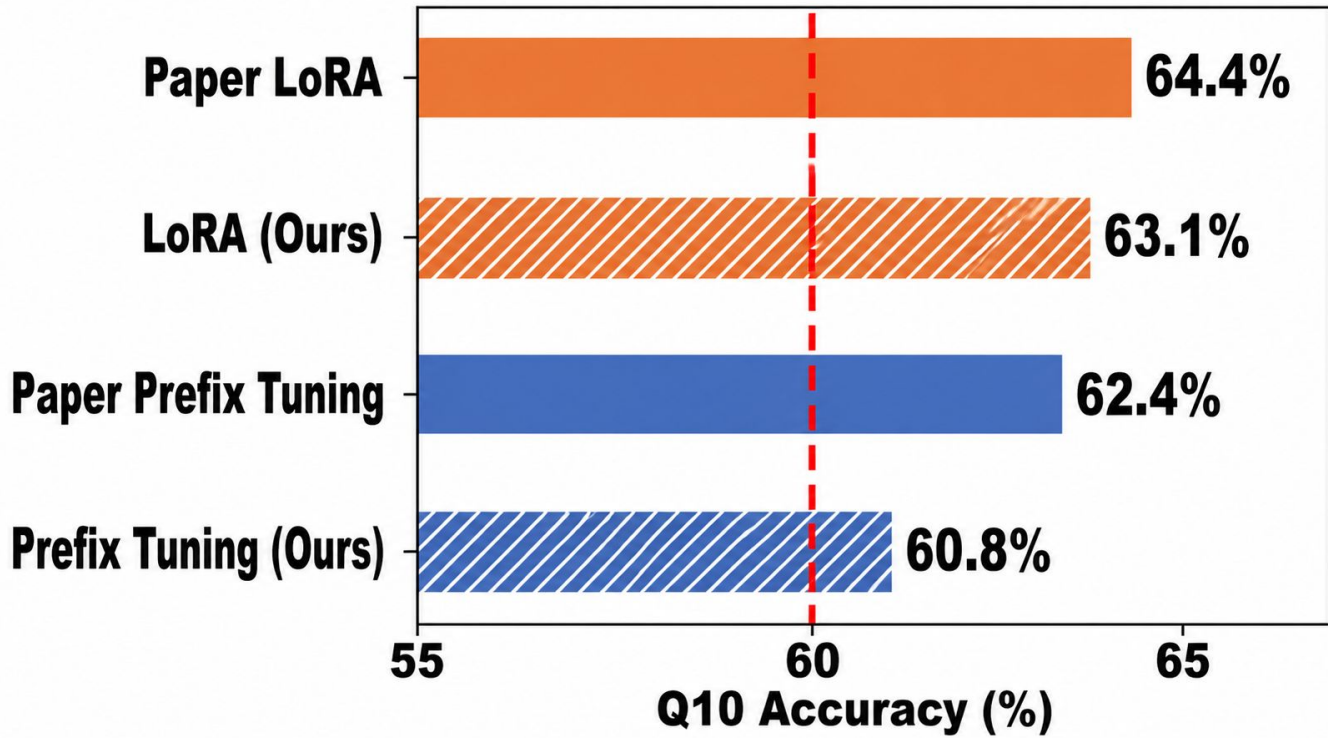
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Fine-Tuning Task

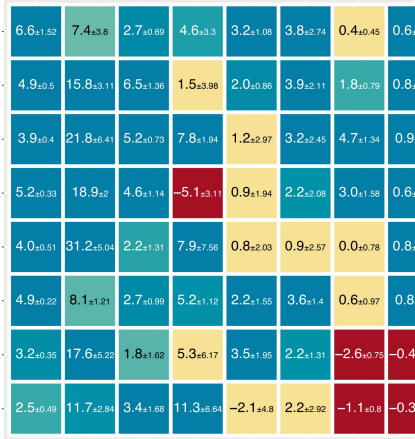


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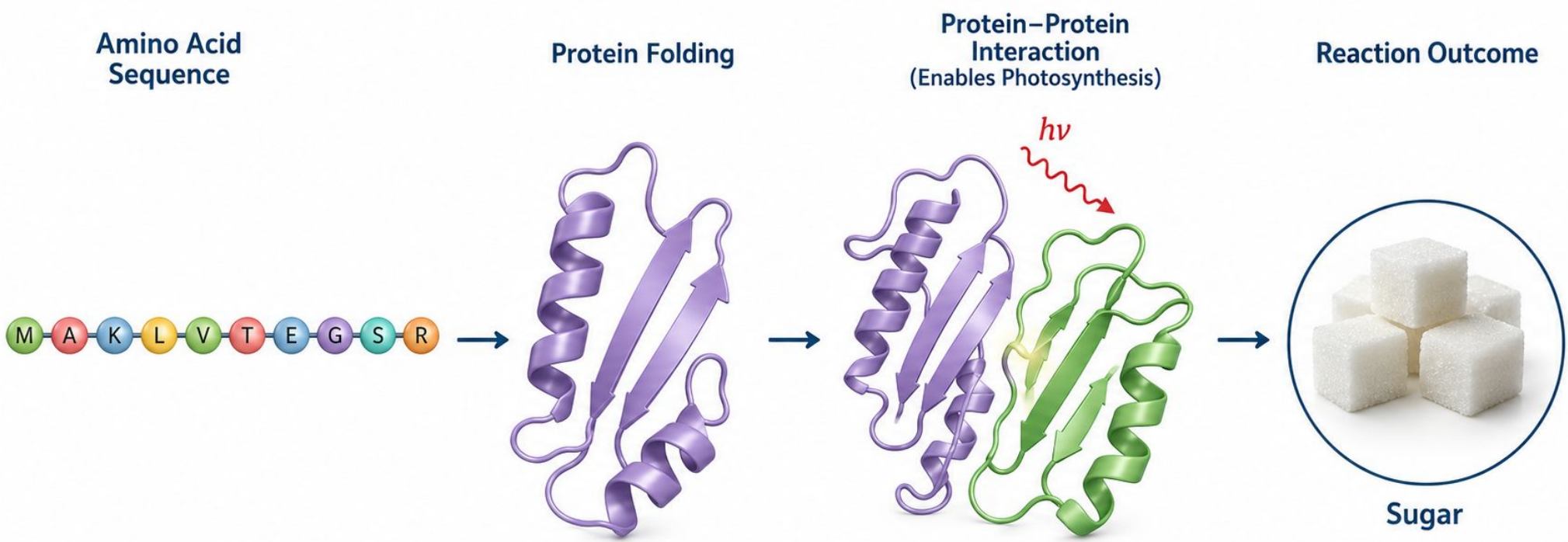
Performance across different fine-tuning schemes. Red line is vanilla ProtT5.

Key takeaway: Always fine-tune your PLMs!



Fine-tuning to play *pin the protein on the cell*

Background



MAKLVTGSR

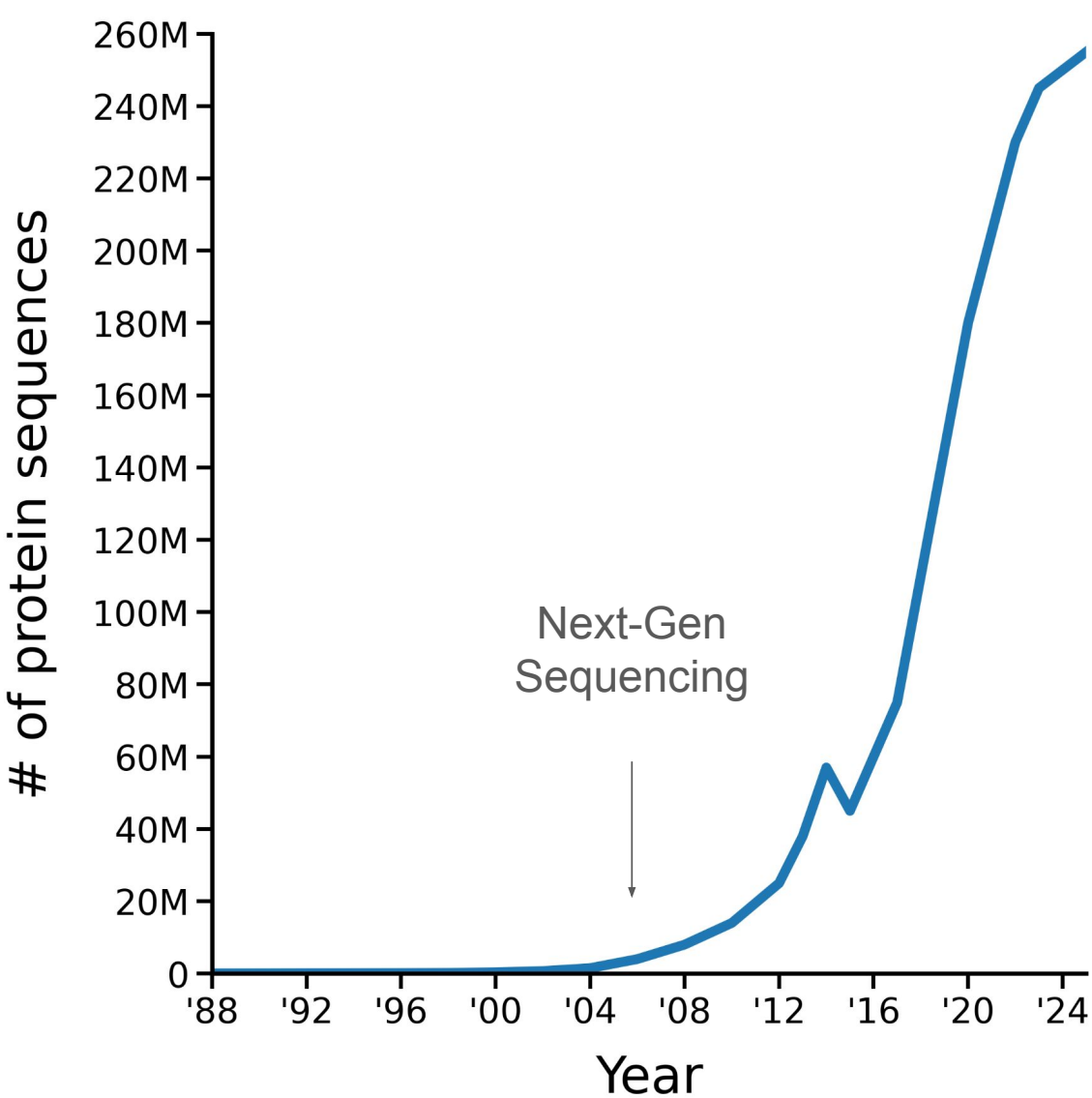
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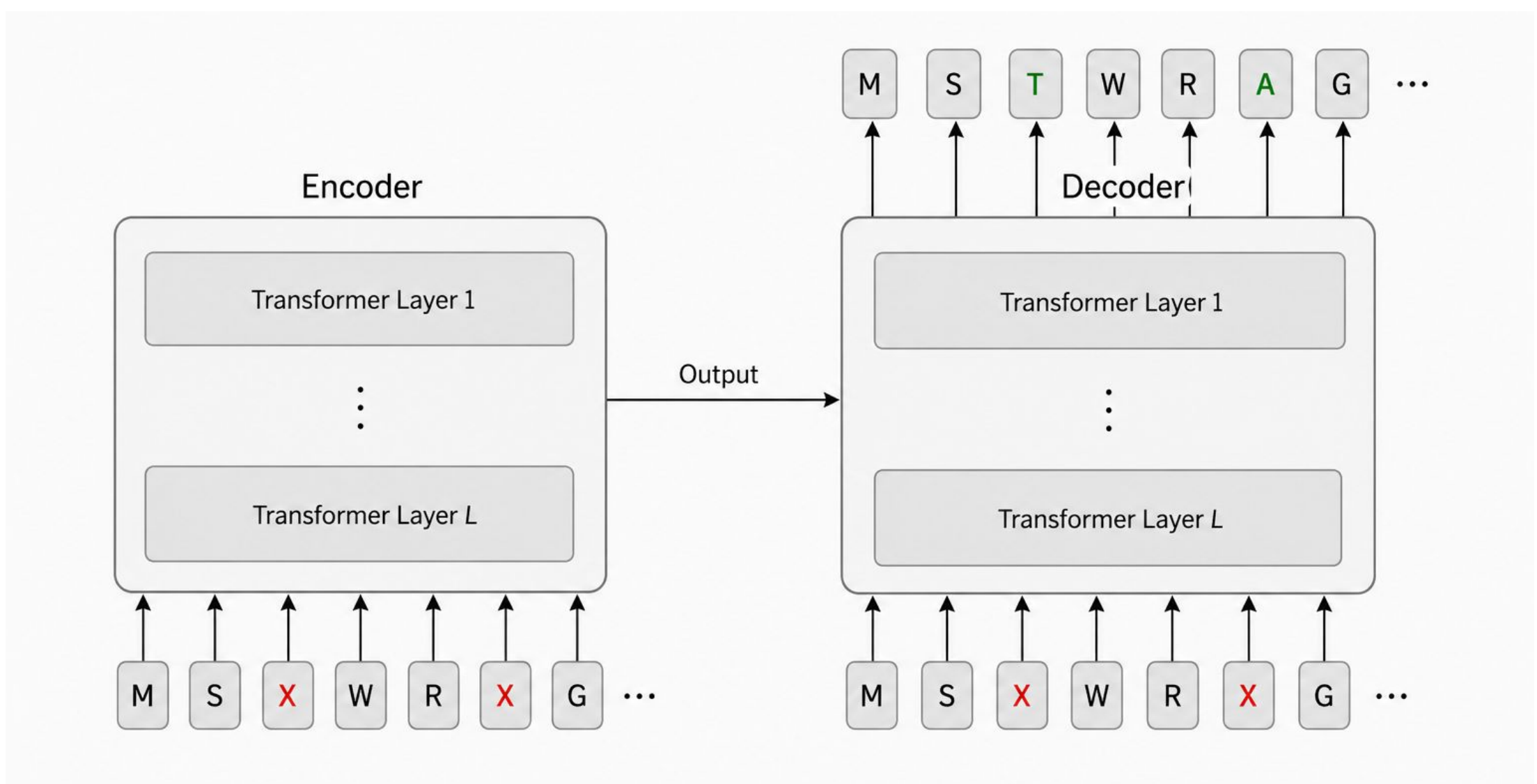
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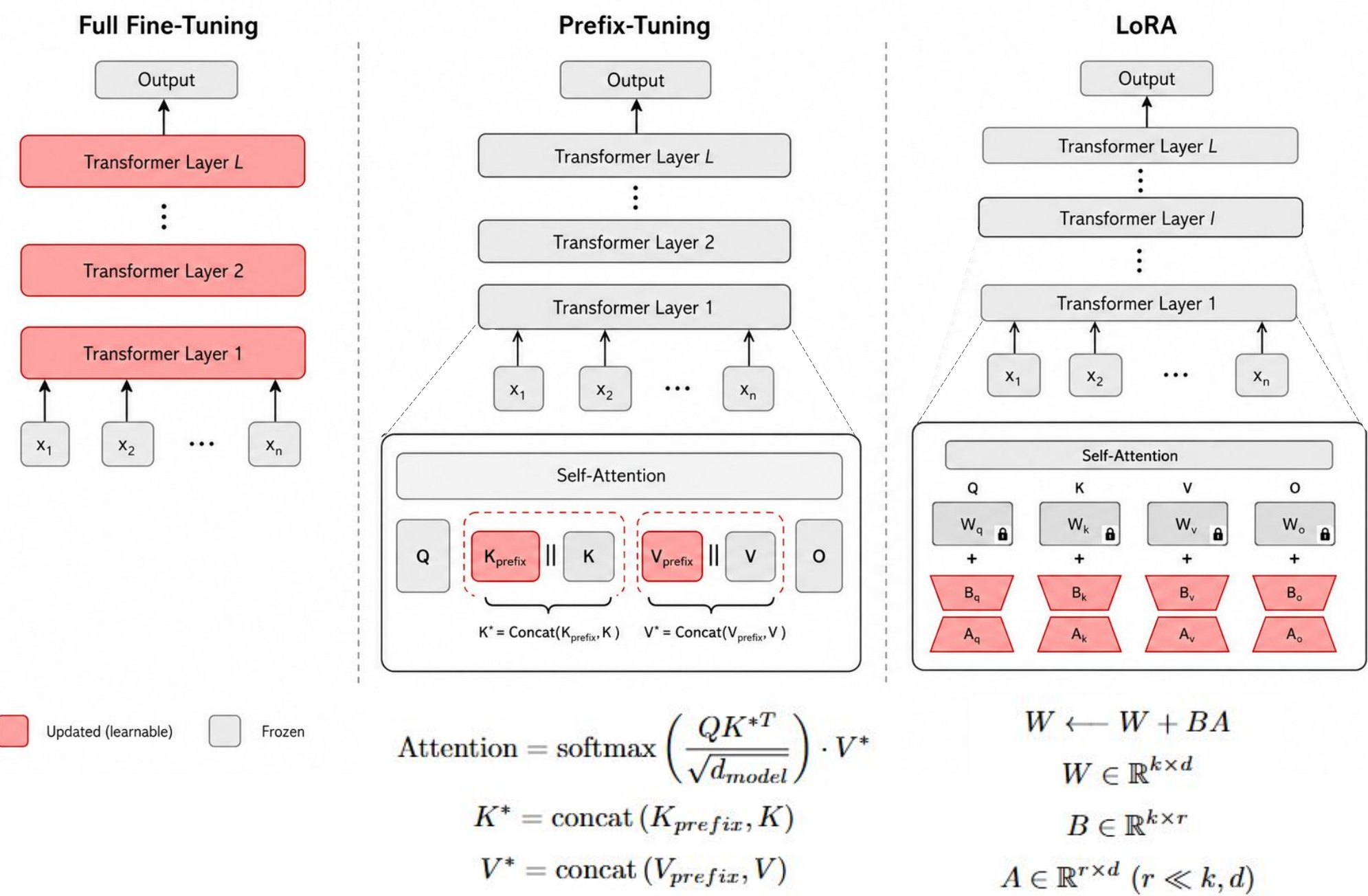
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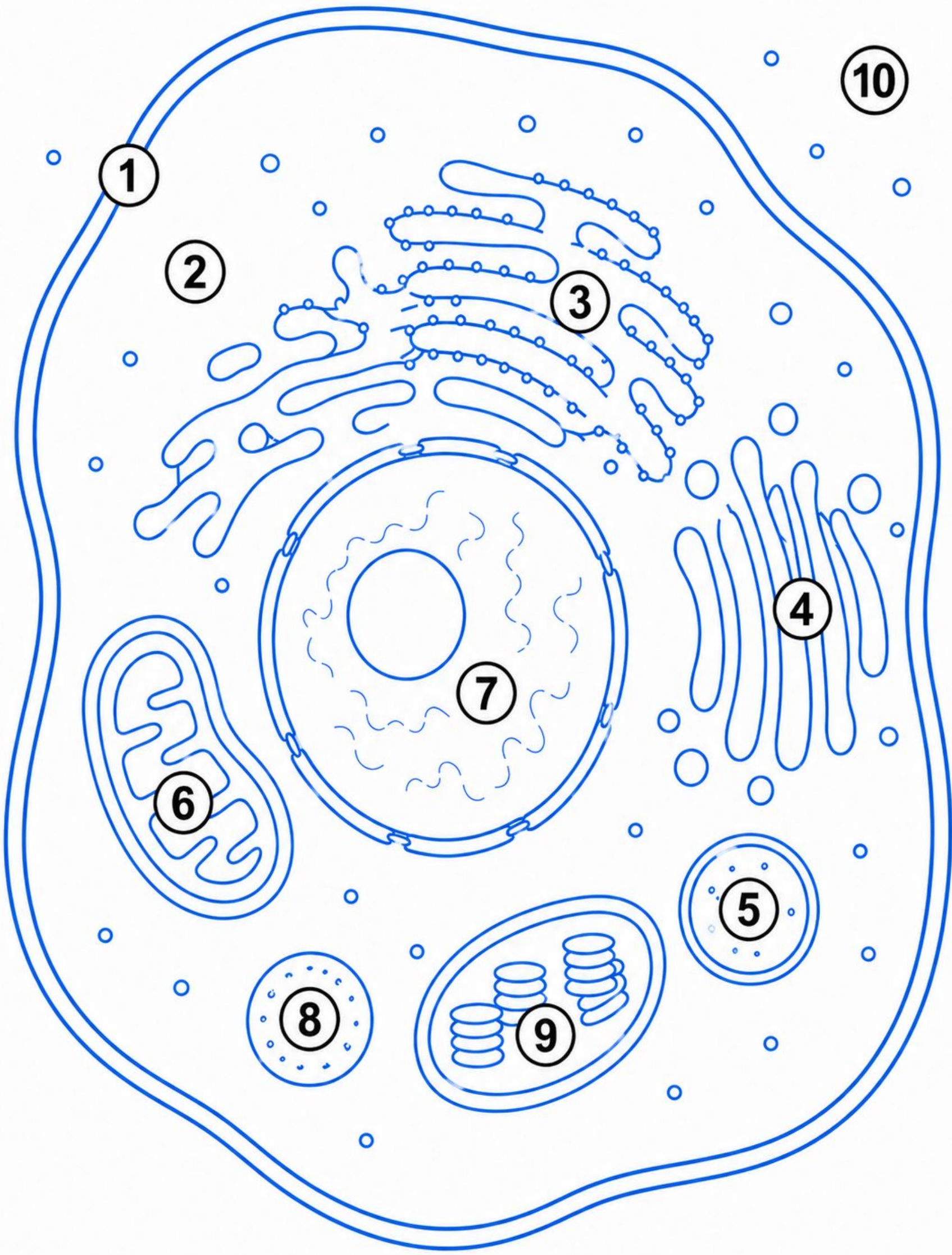


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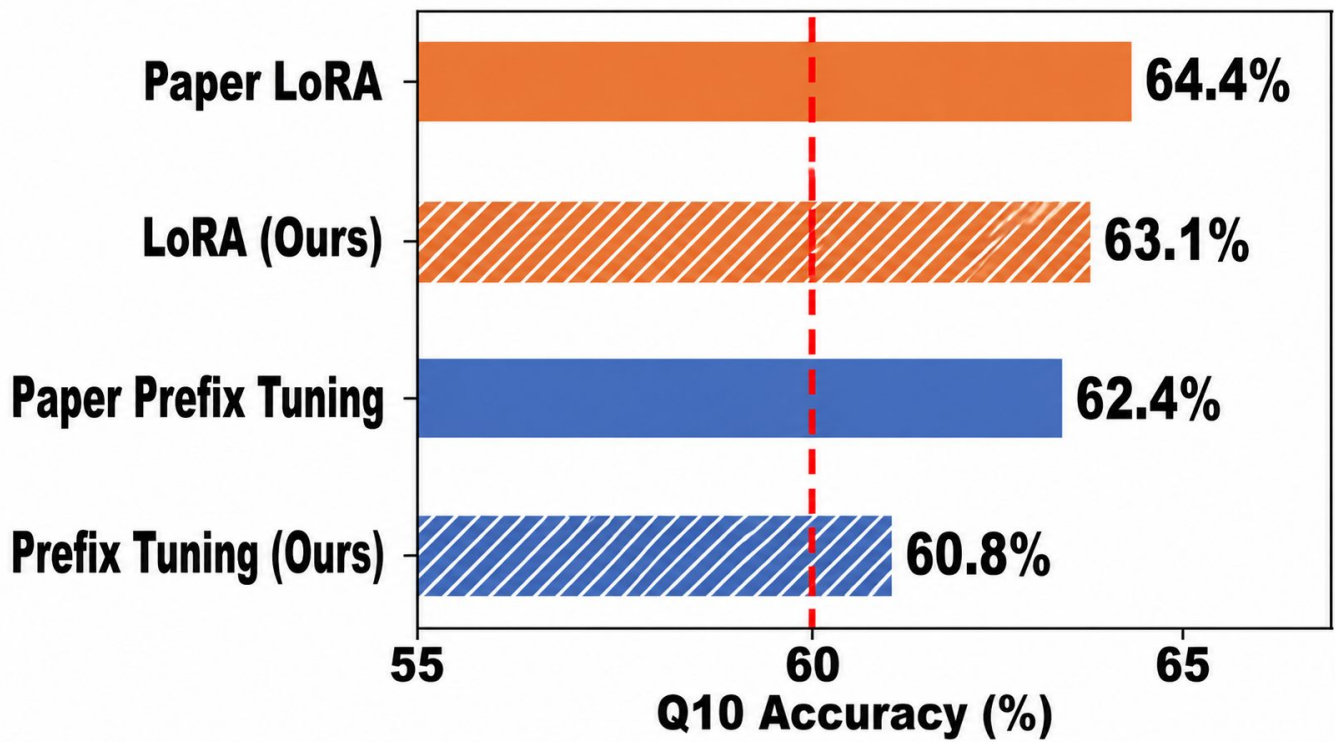
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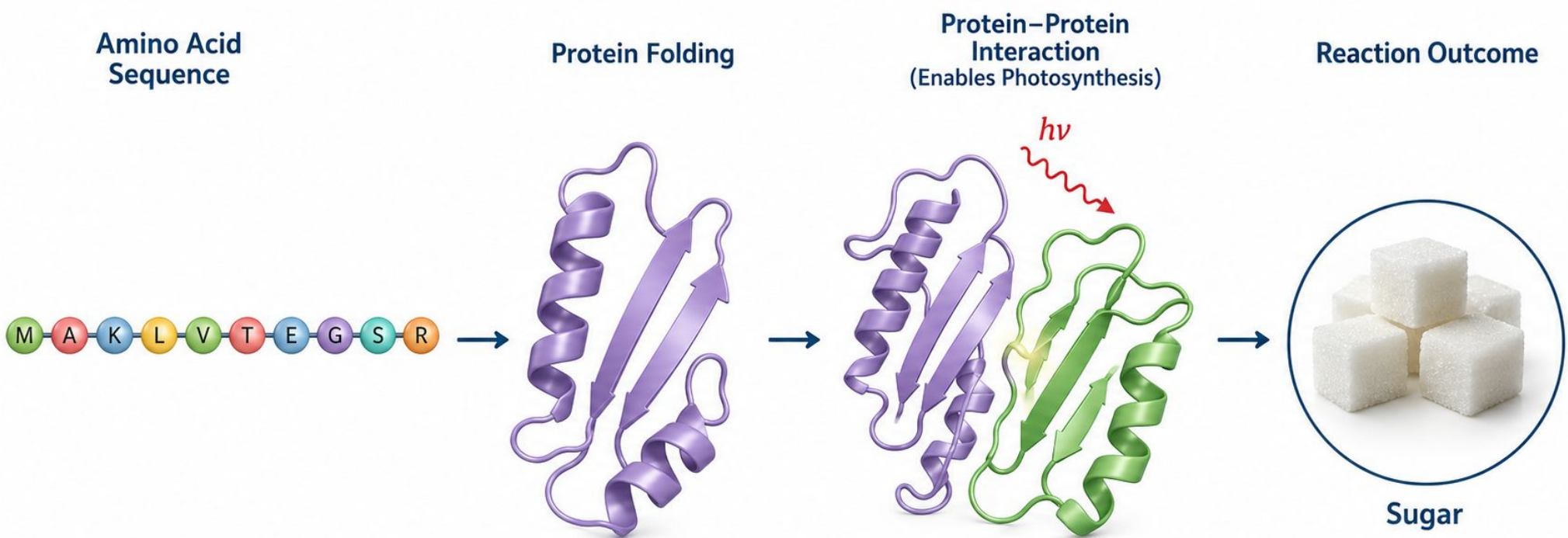


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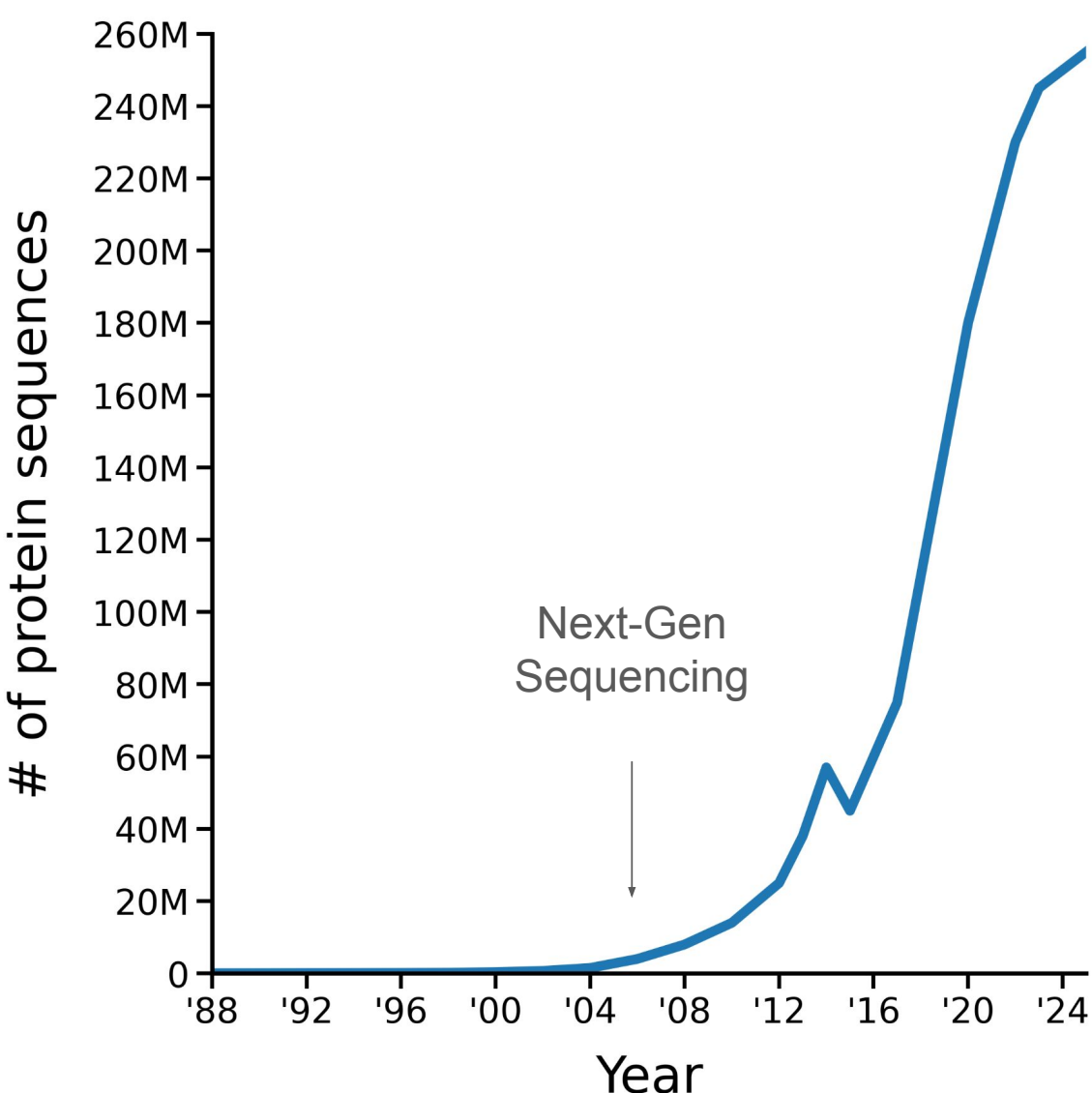
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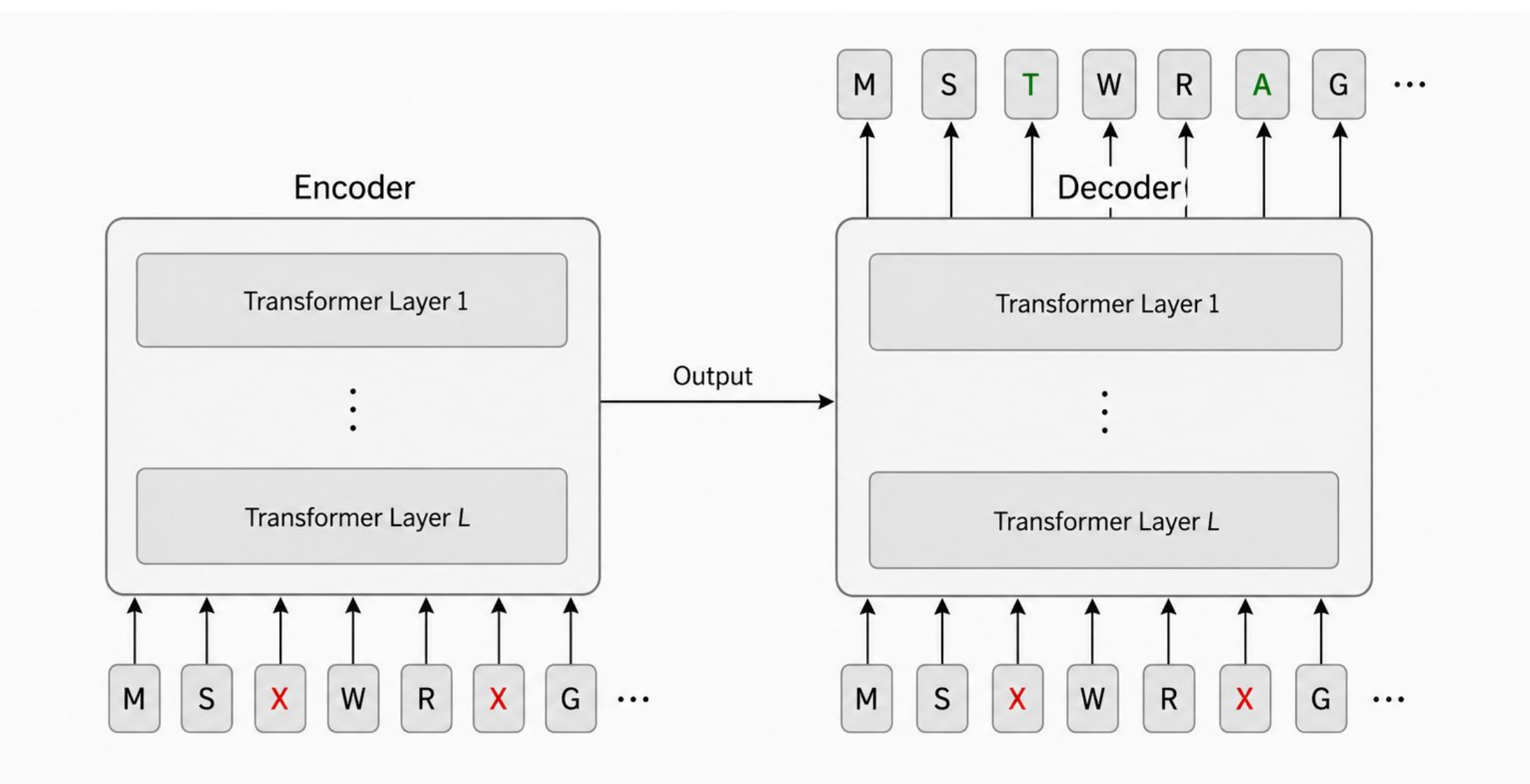
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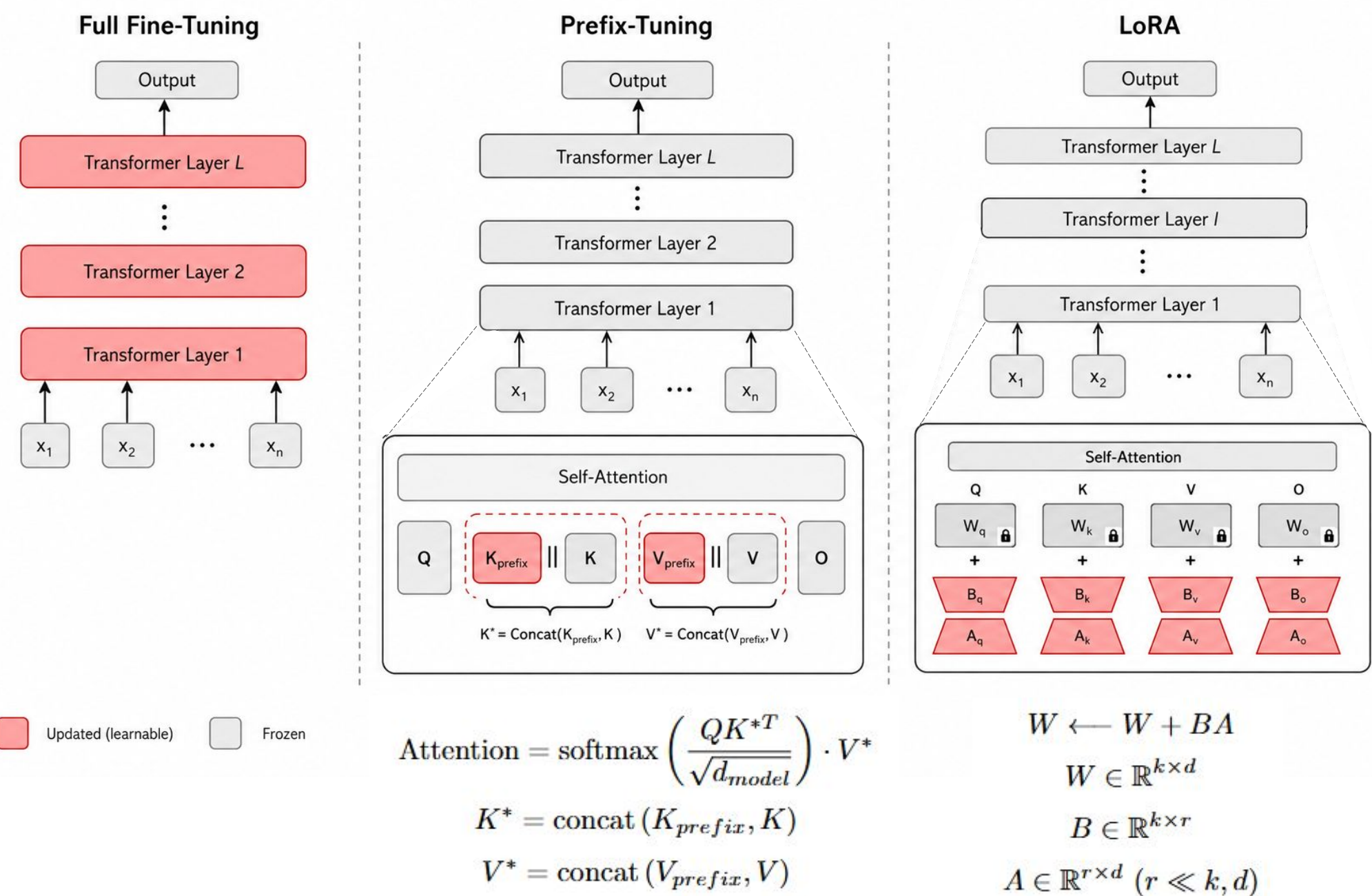
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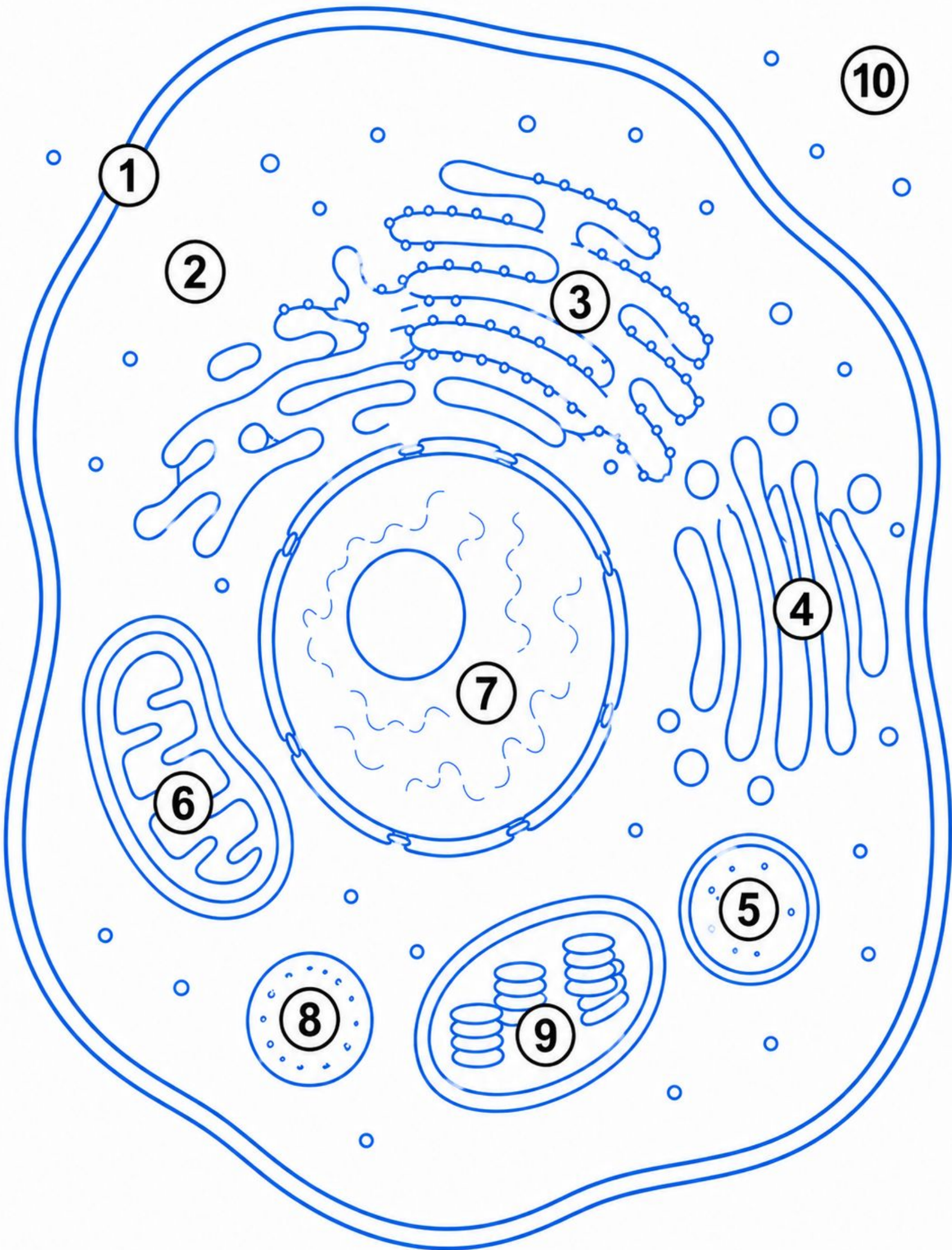


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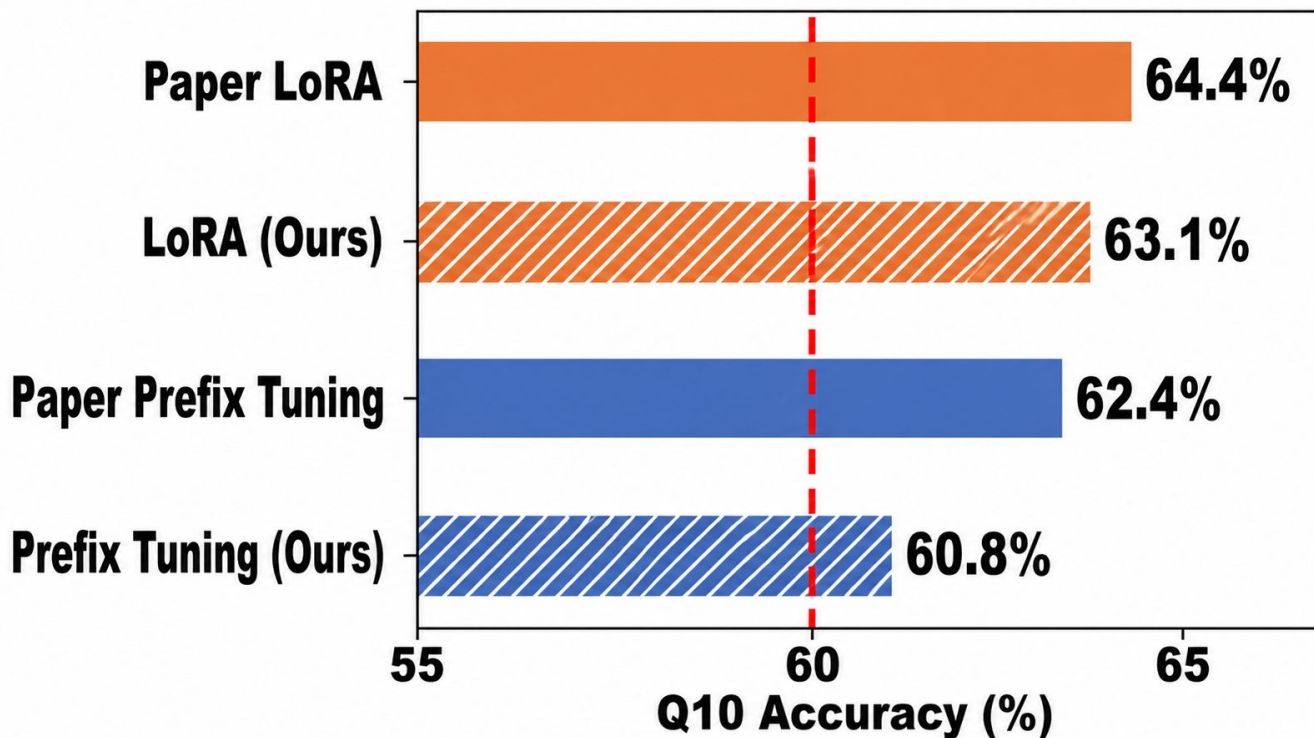
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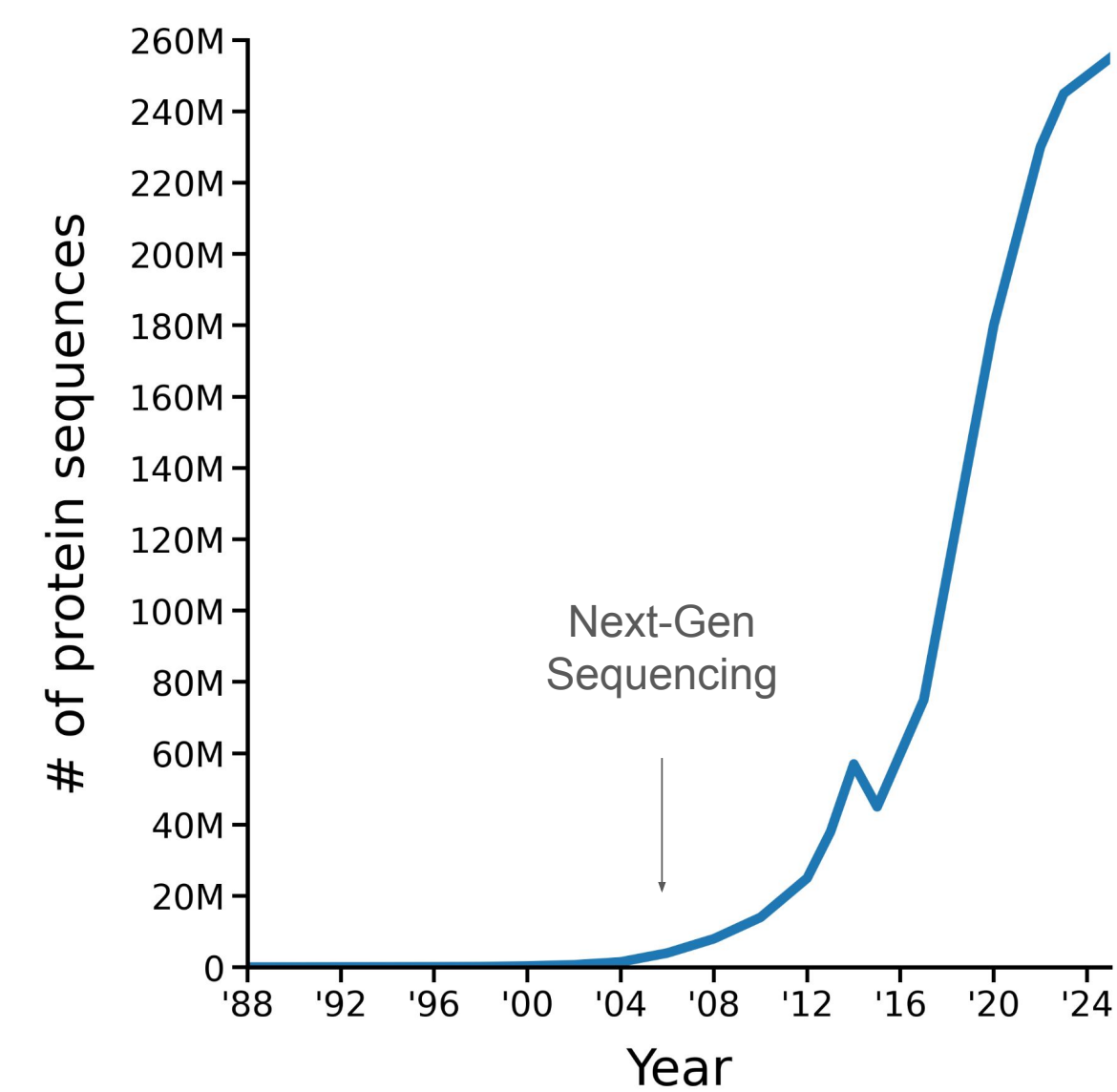
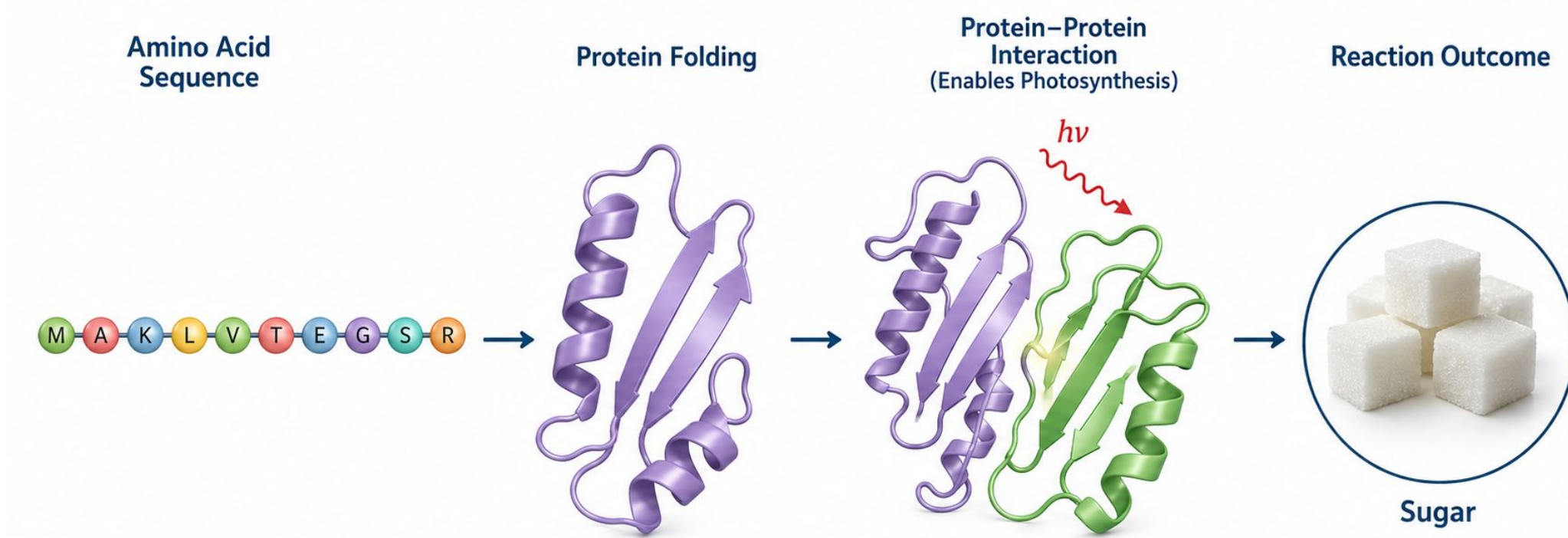


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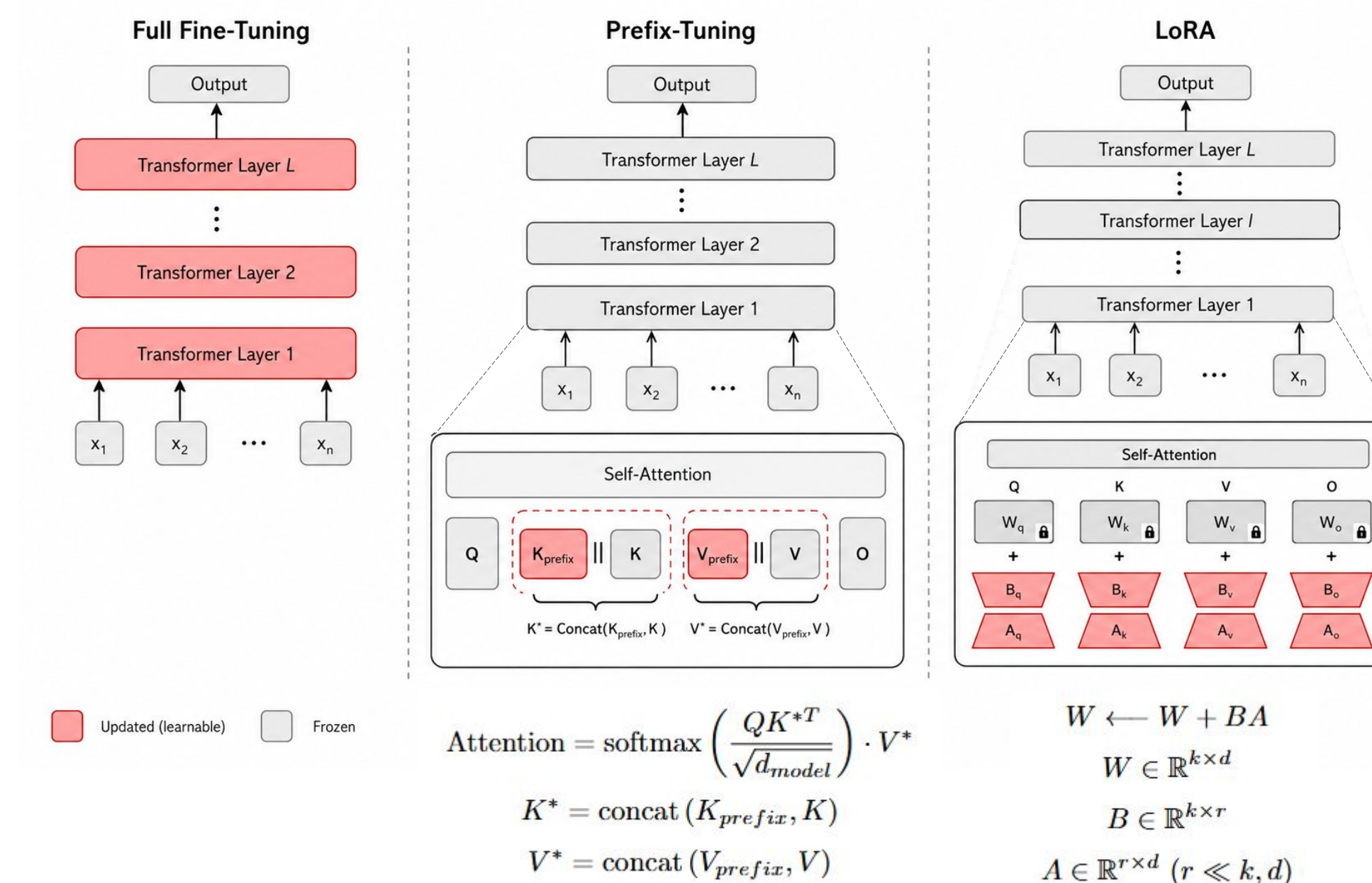
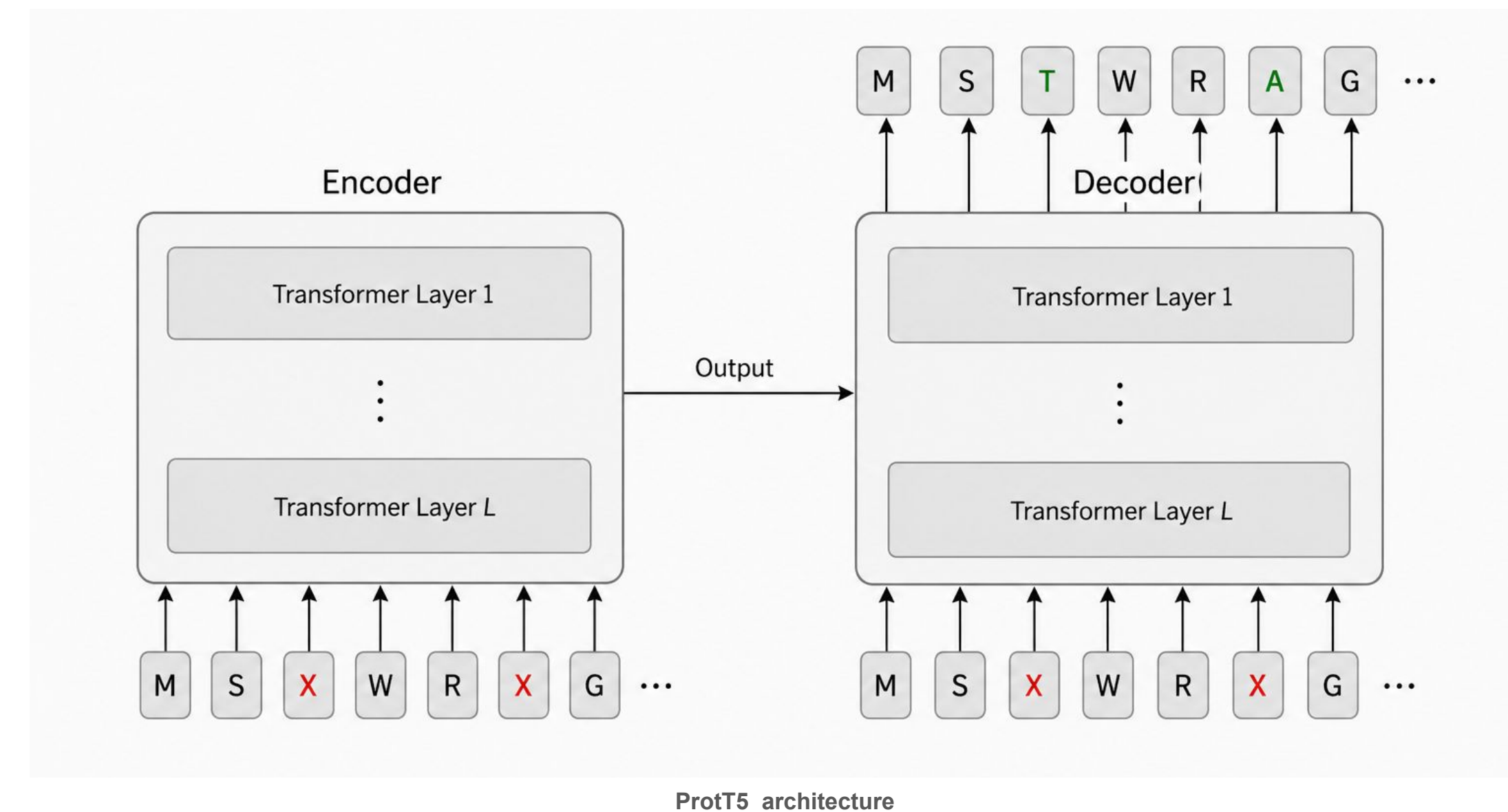
Fine-tuning a protein language model to play *pin the protein on the cell*

Background



Unlabeled protein sequence data is abundant. Protein sequences in UniProt database over the last few decades

Architecture

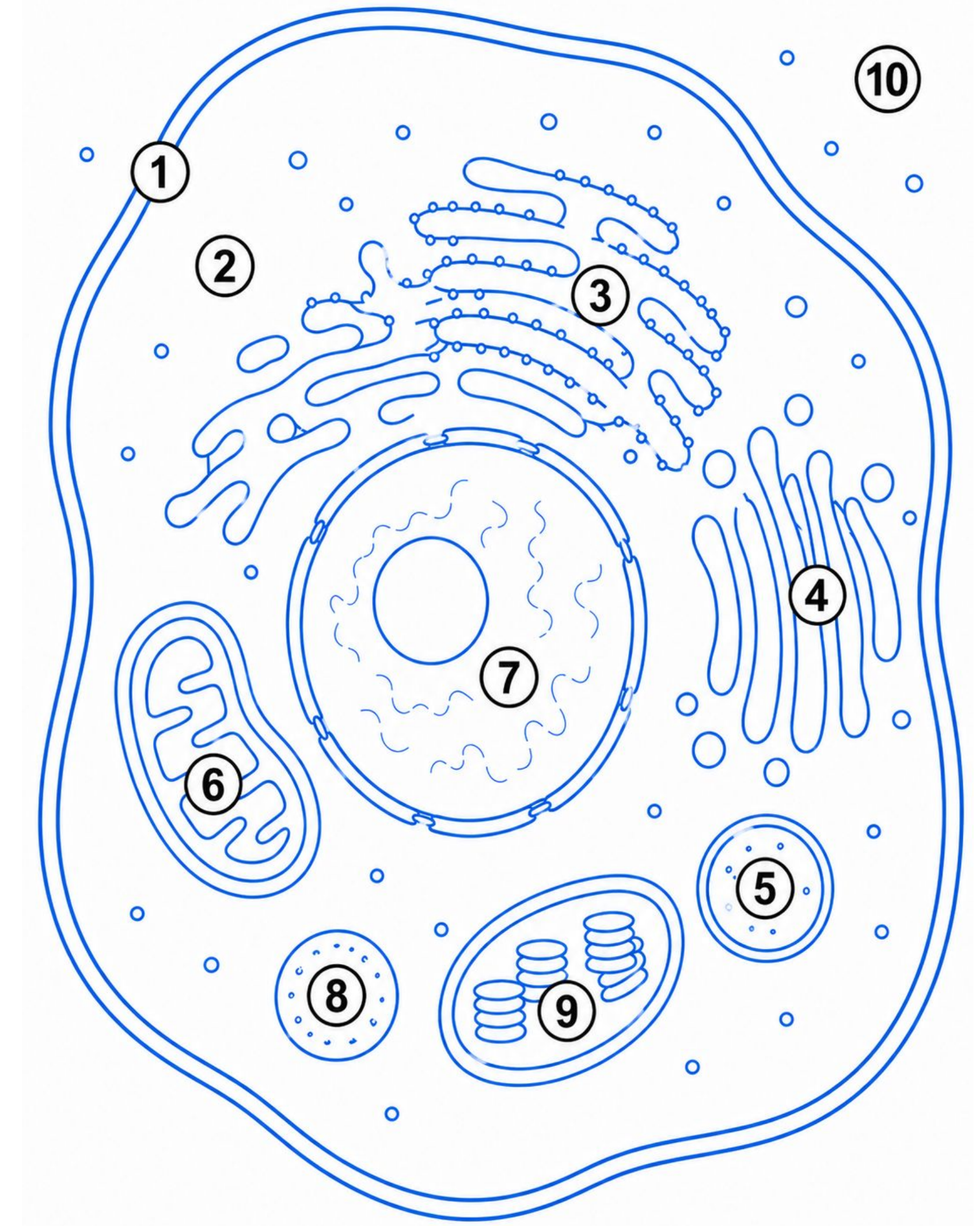


Schematics of Full Fine-Tuning, Prefix-Tuning and Low-Rank Adaptation (LoRA)

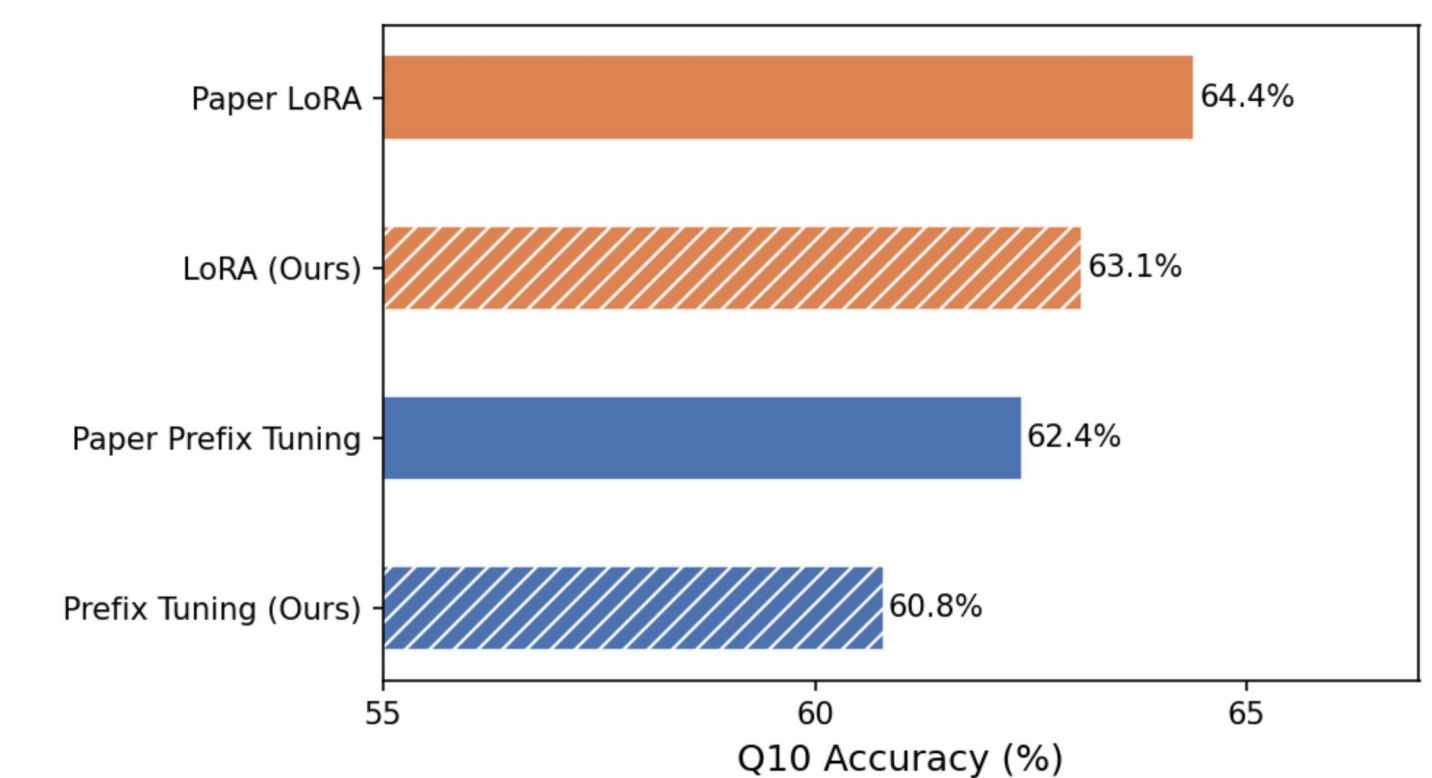
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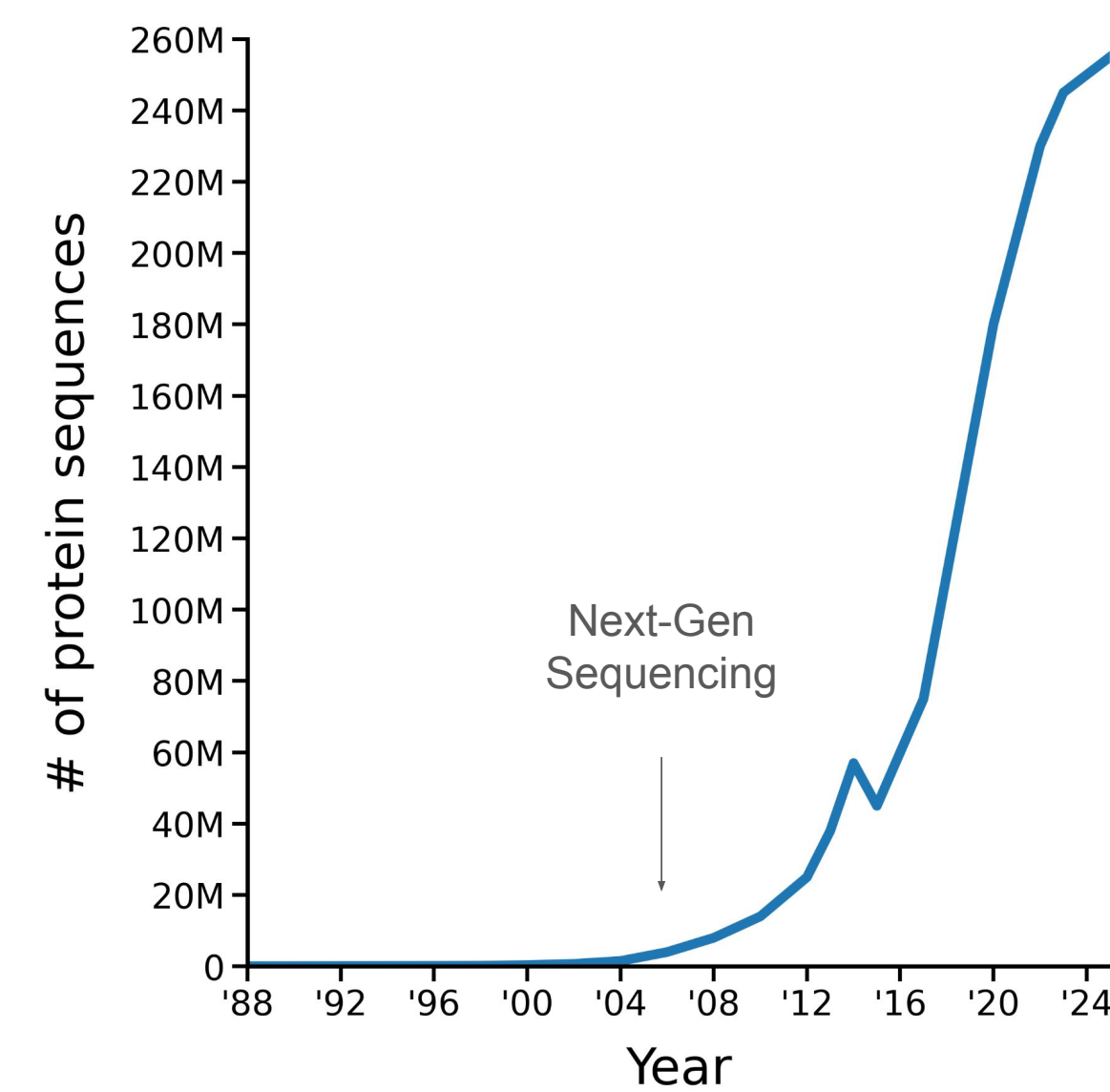
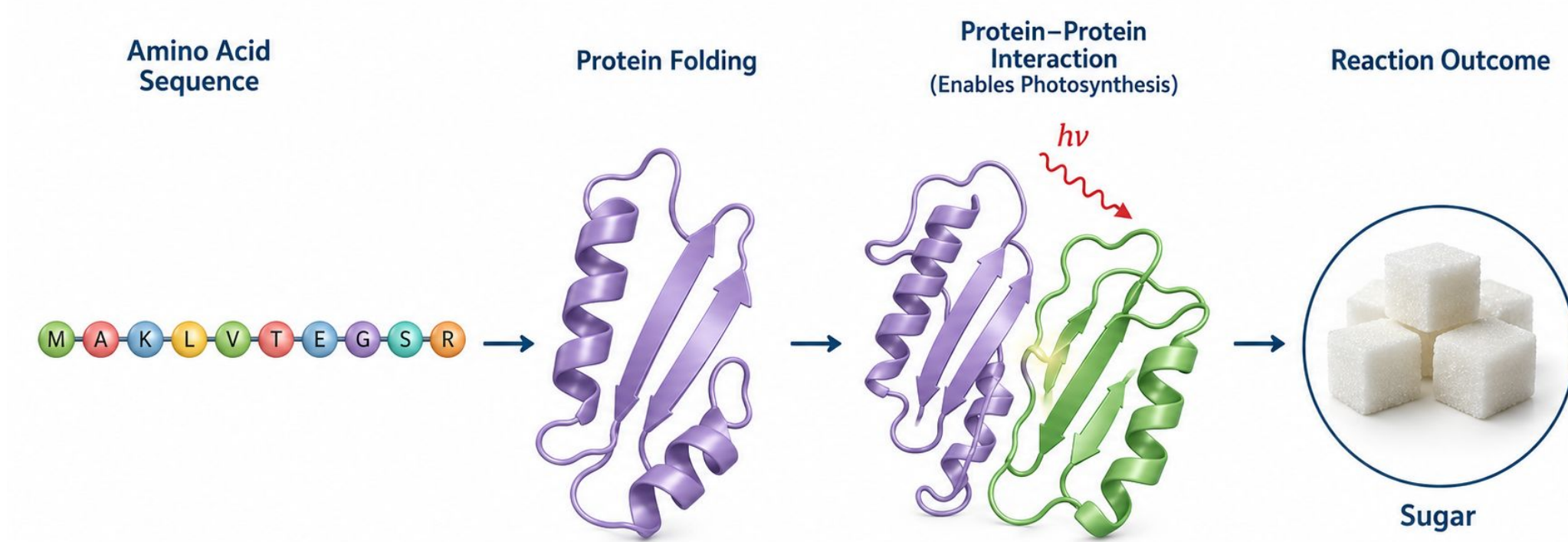


1. Cell Membrane
2. Cytoplasm
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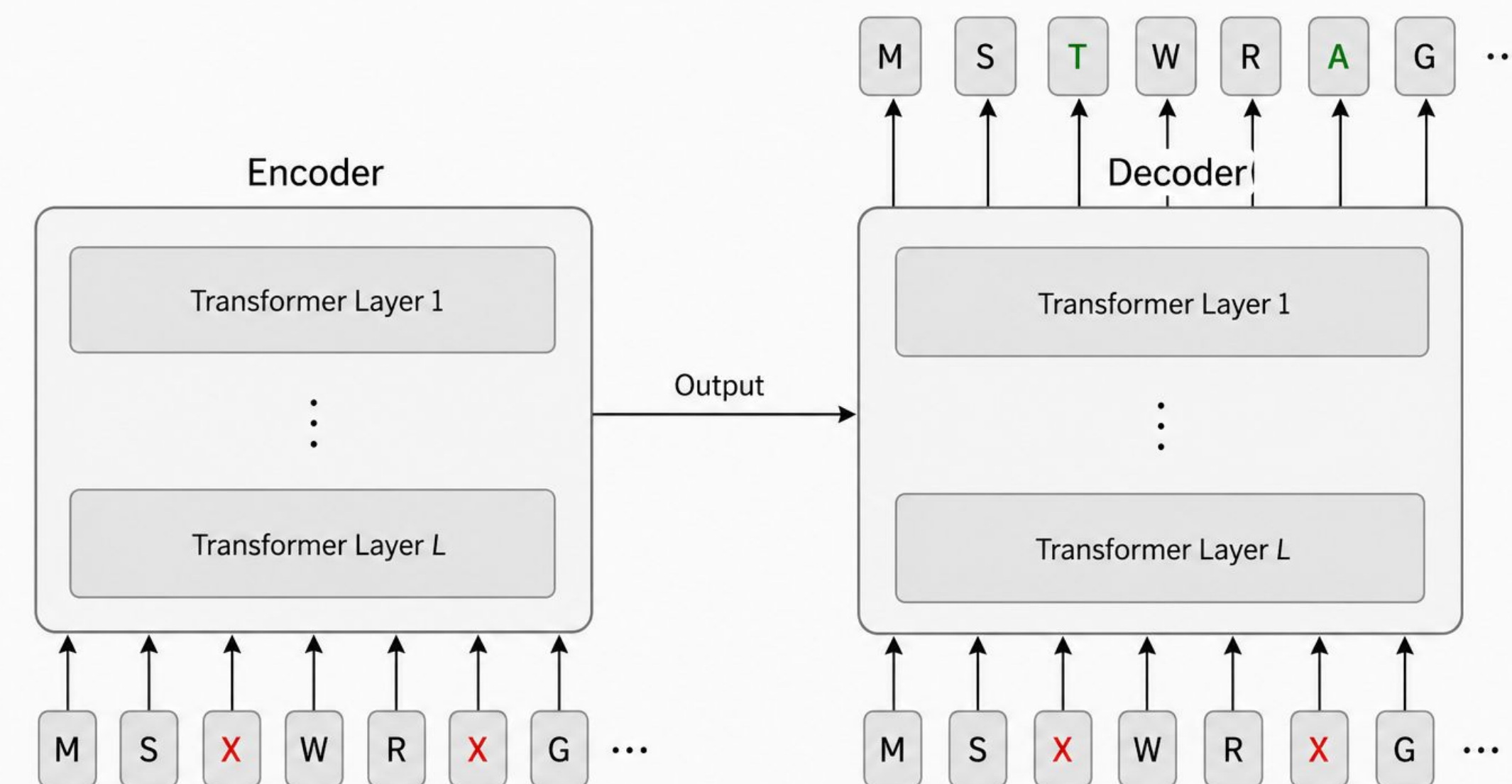
Performance comparison across different schemes, measured in Q10 accuracy

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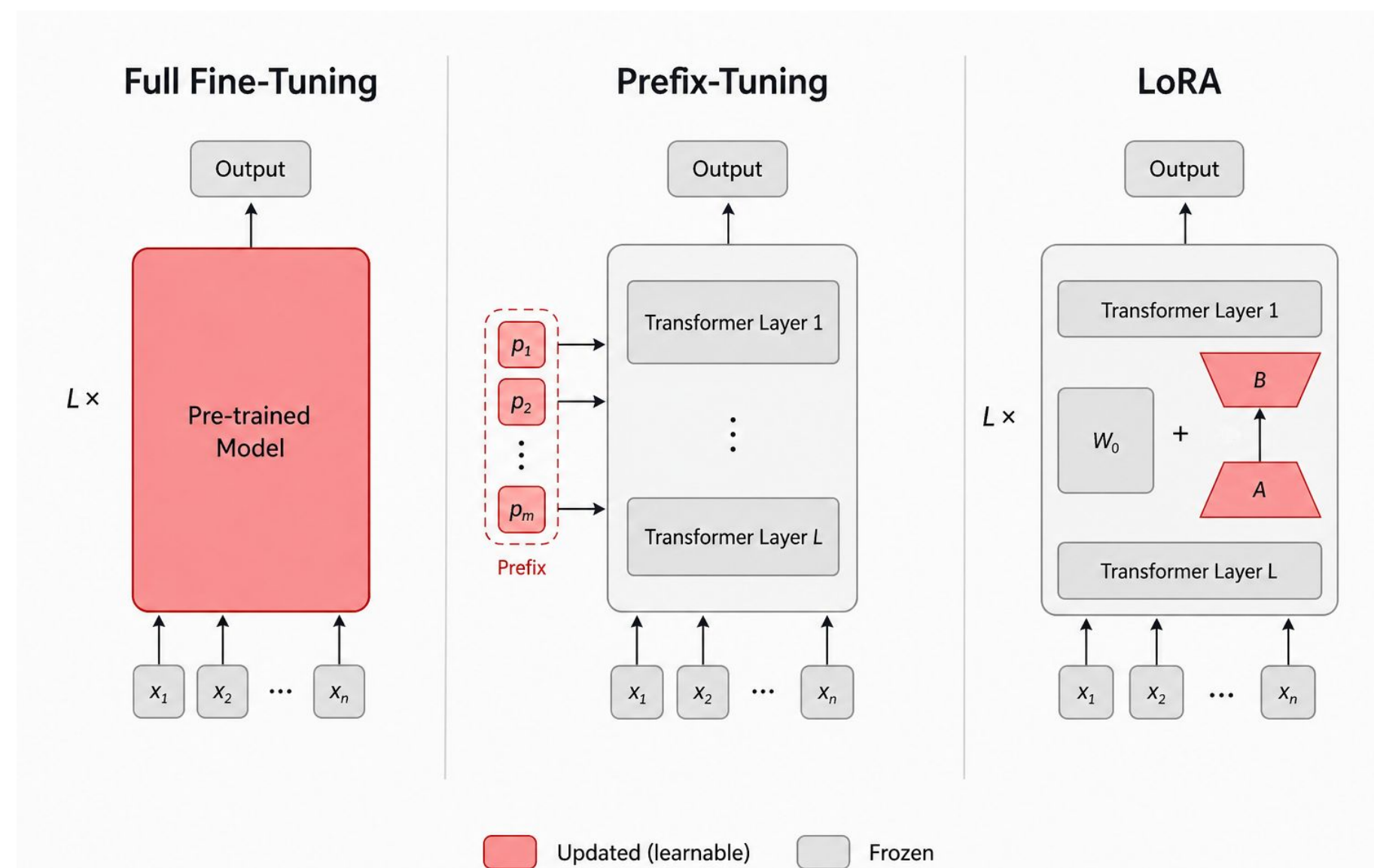
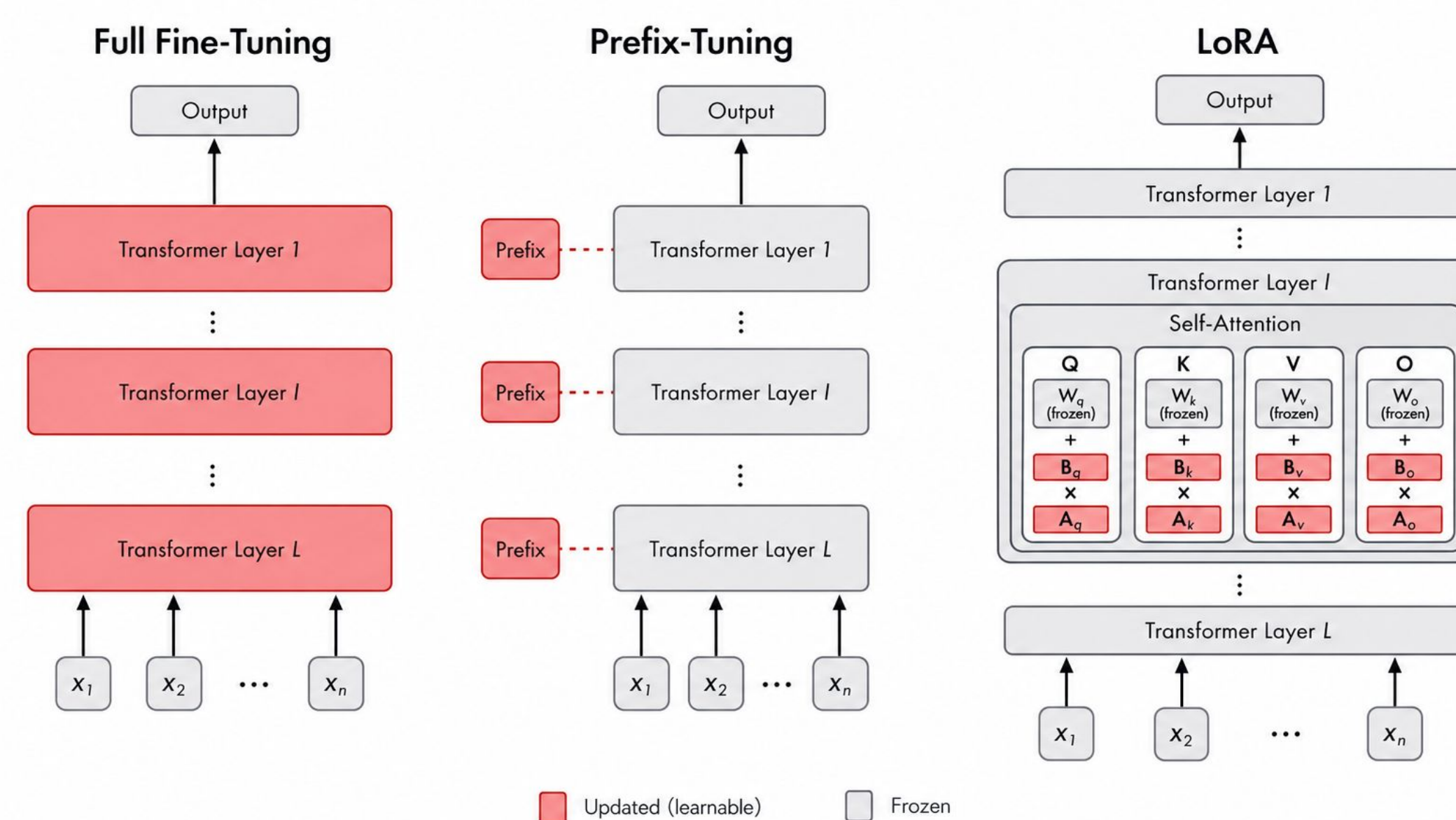


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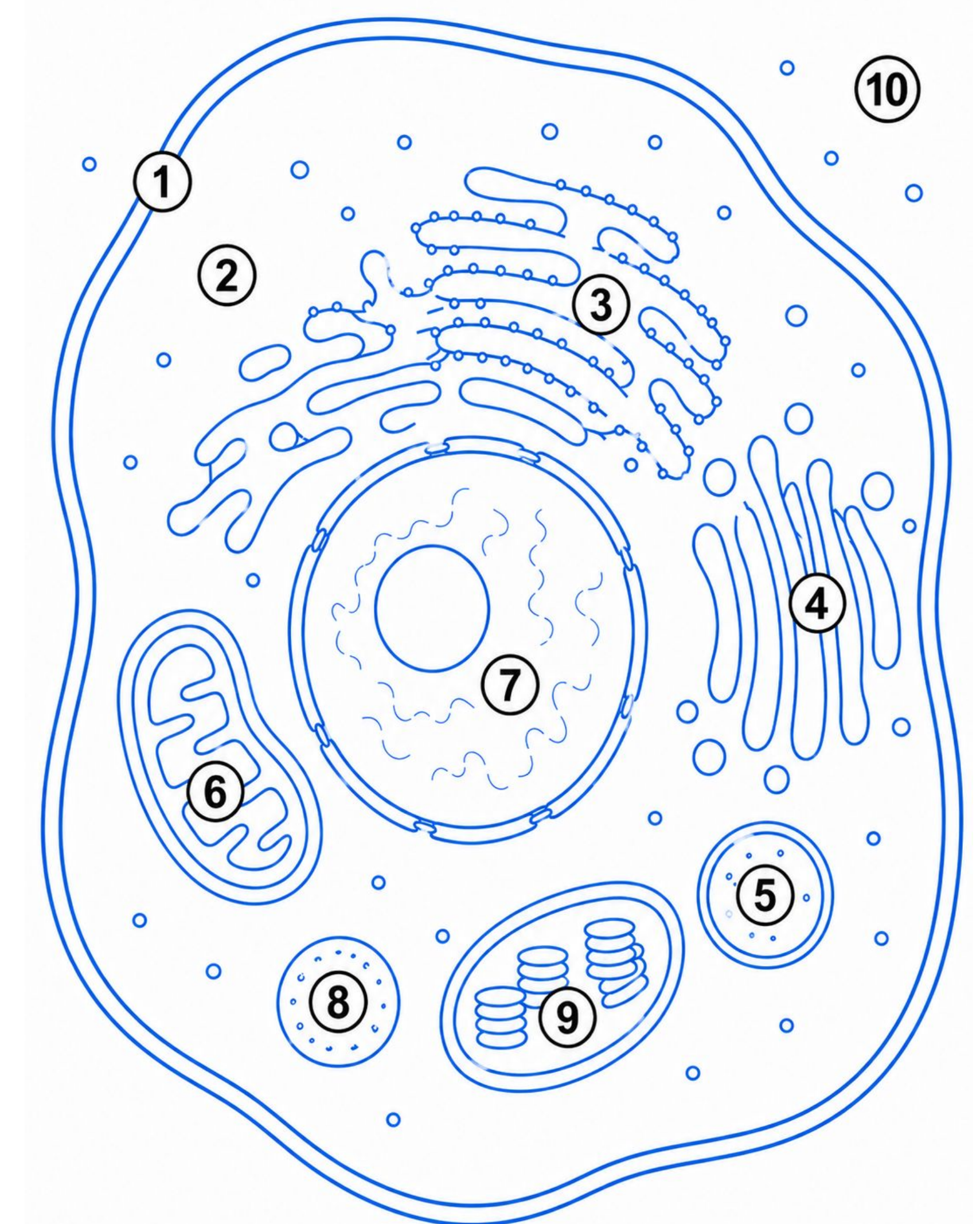
Labeled



ProtT5 architecture

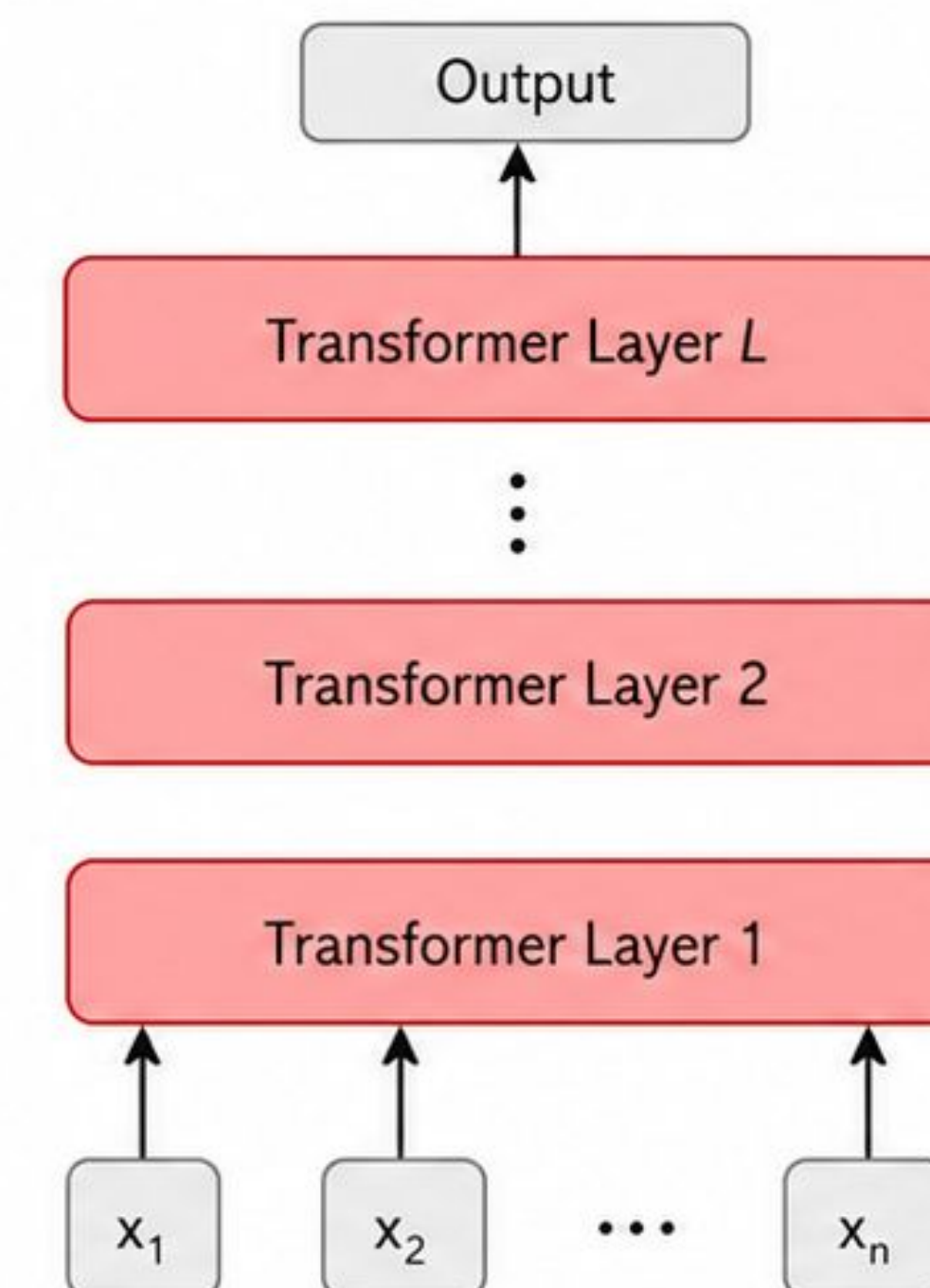


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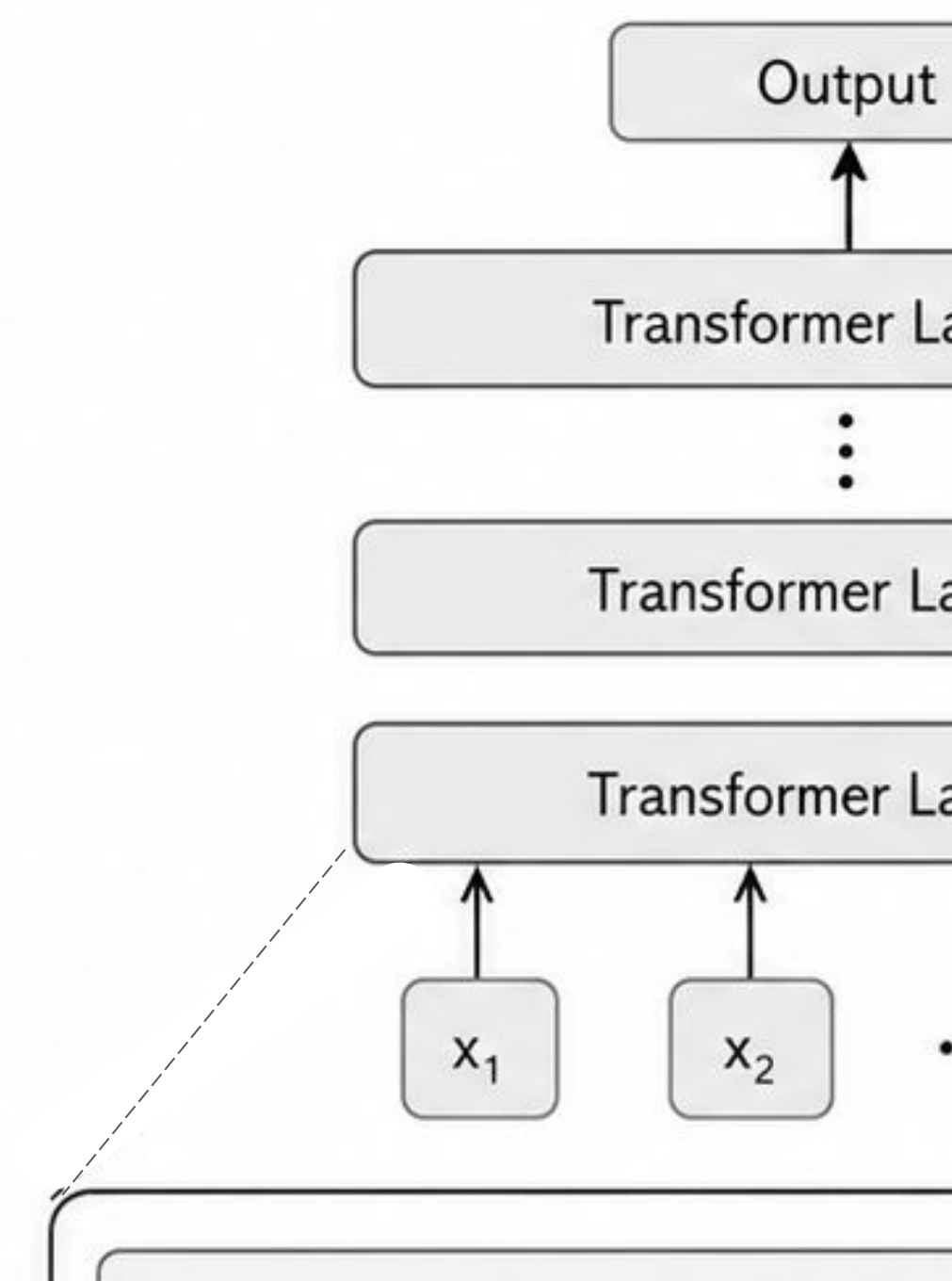


1. Cell Membrane

Full Fine-Tuning

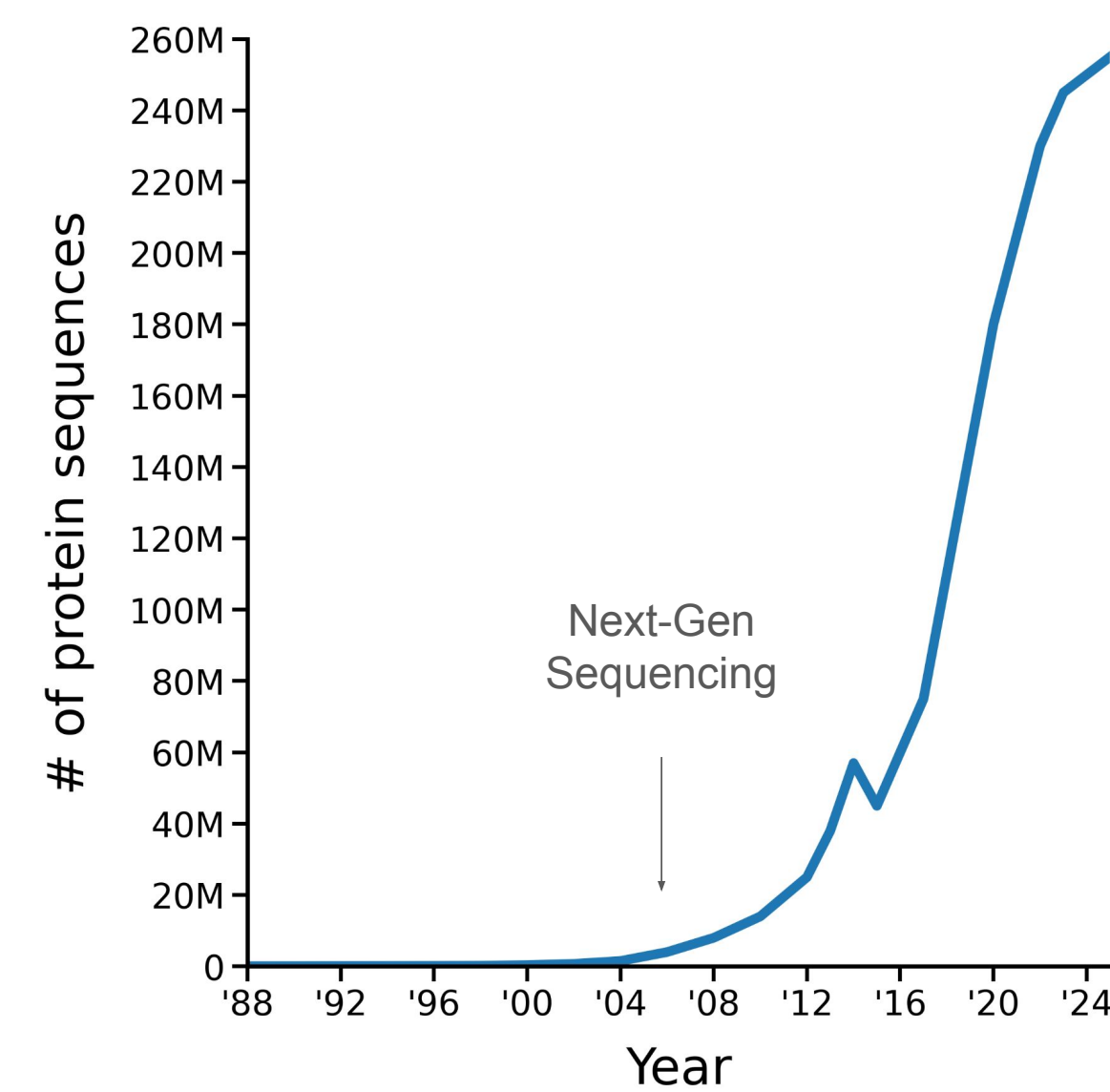


Prefix-Tuning



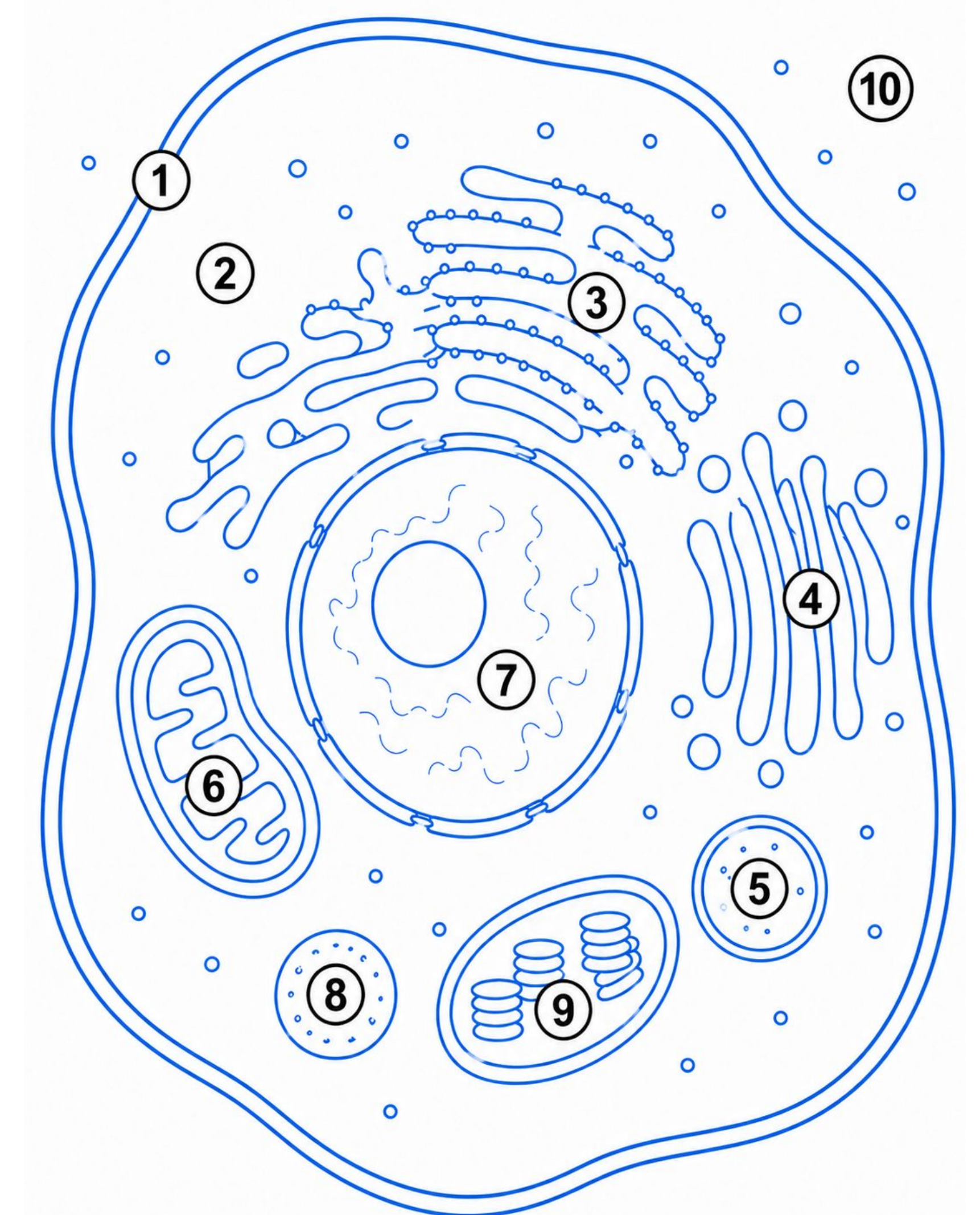
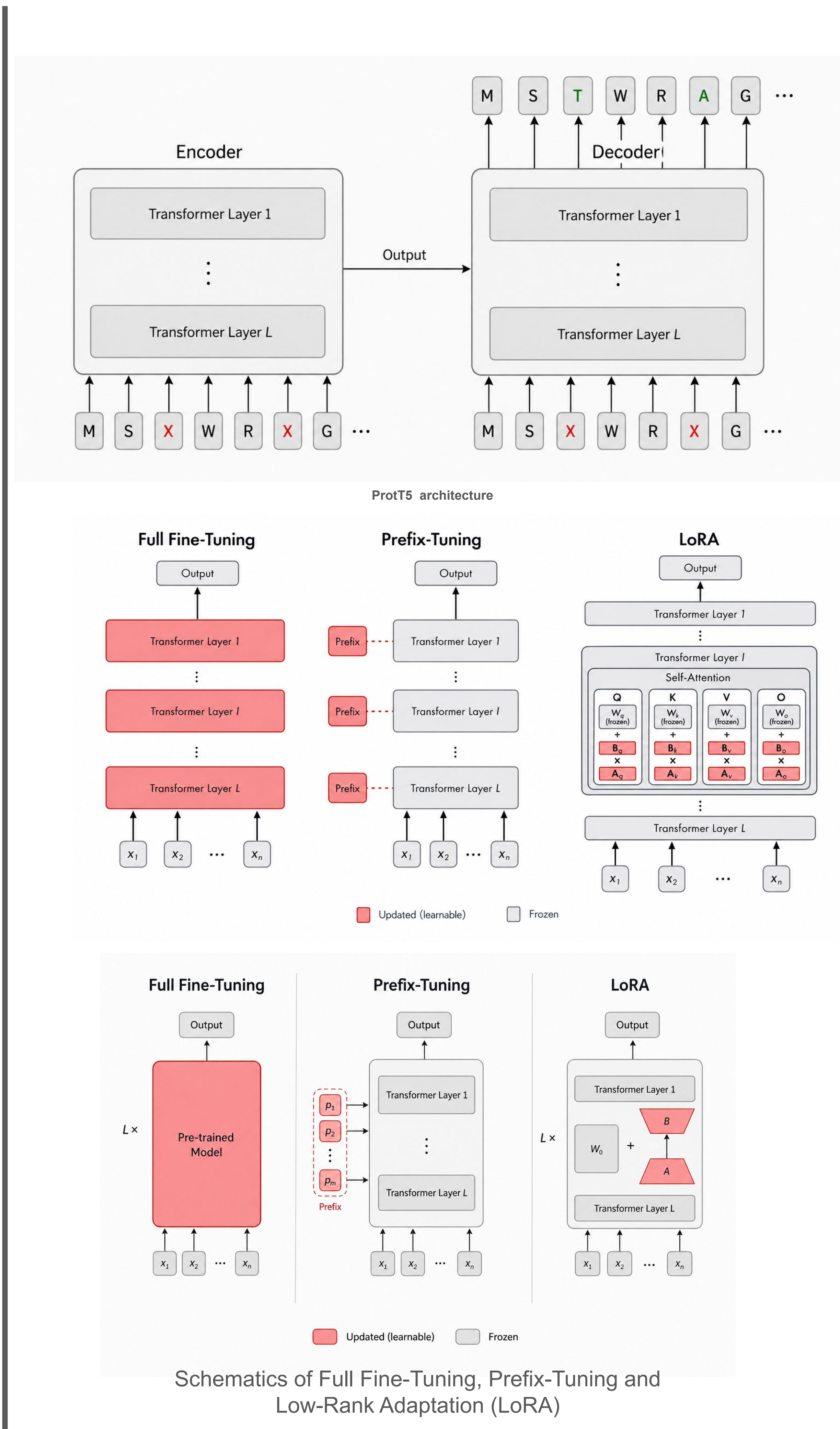
Self-Attention

Fine-tuning a protein language model to play *pin the protein on the cell*

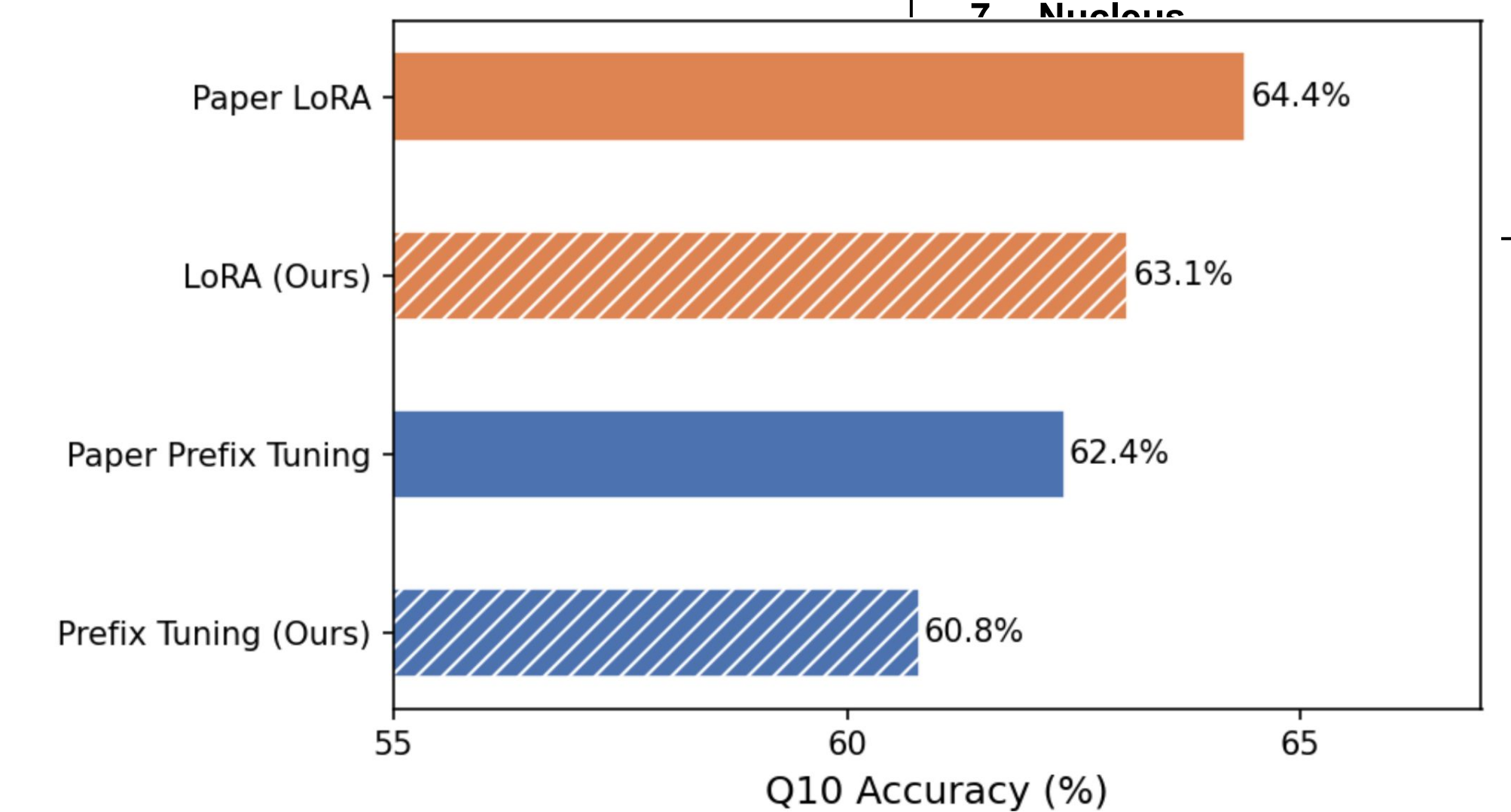


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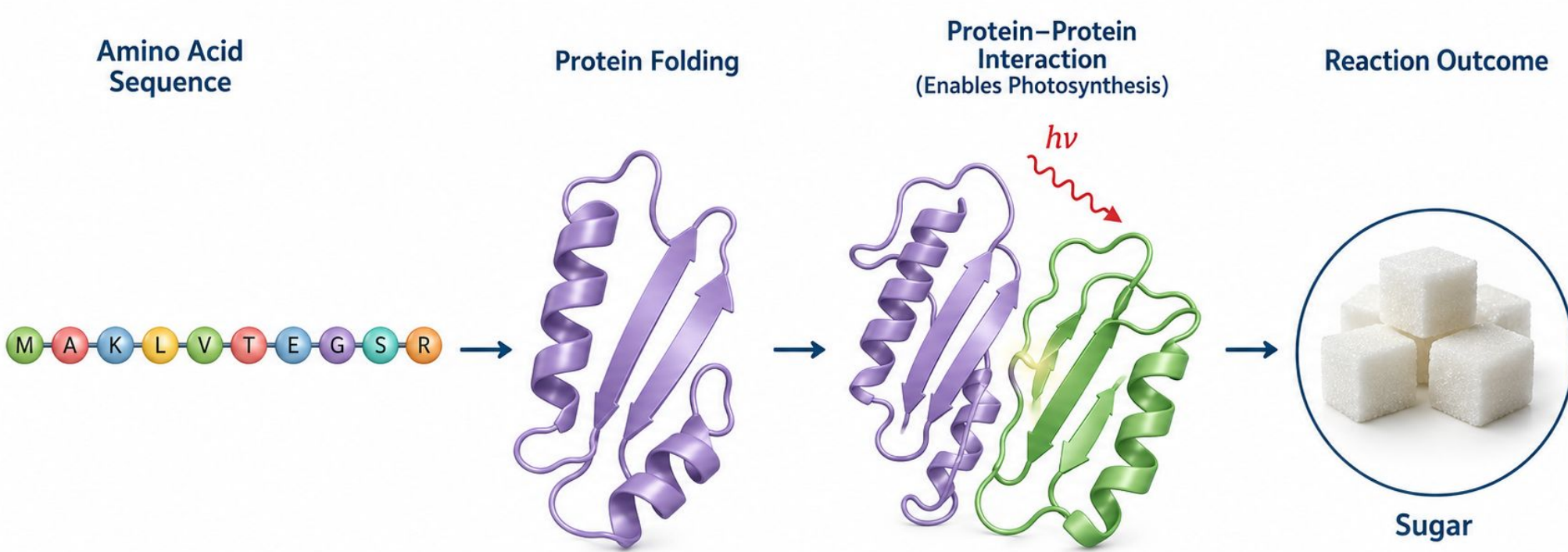
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Performance comparison across different schemes, measured in Q10 accuracy

Fine-tuning a protein language model to play *pin the protein on the cell*

Background



MAKLVTGSR

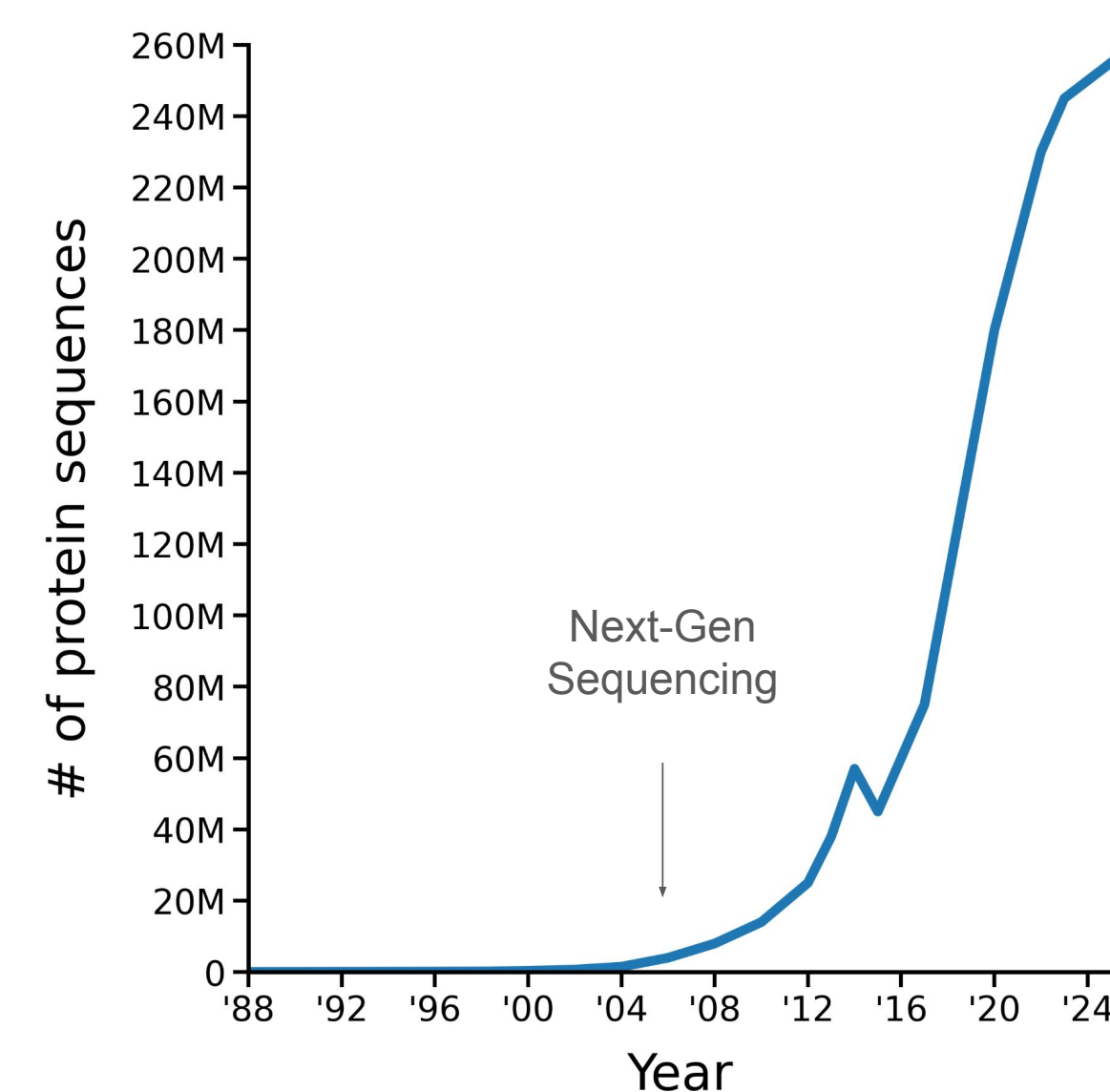
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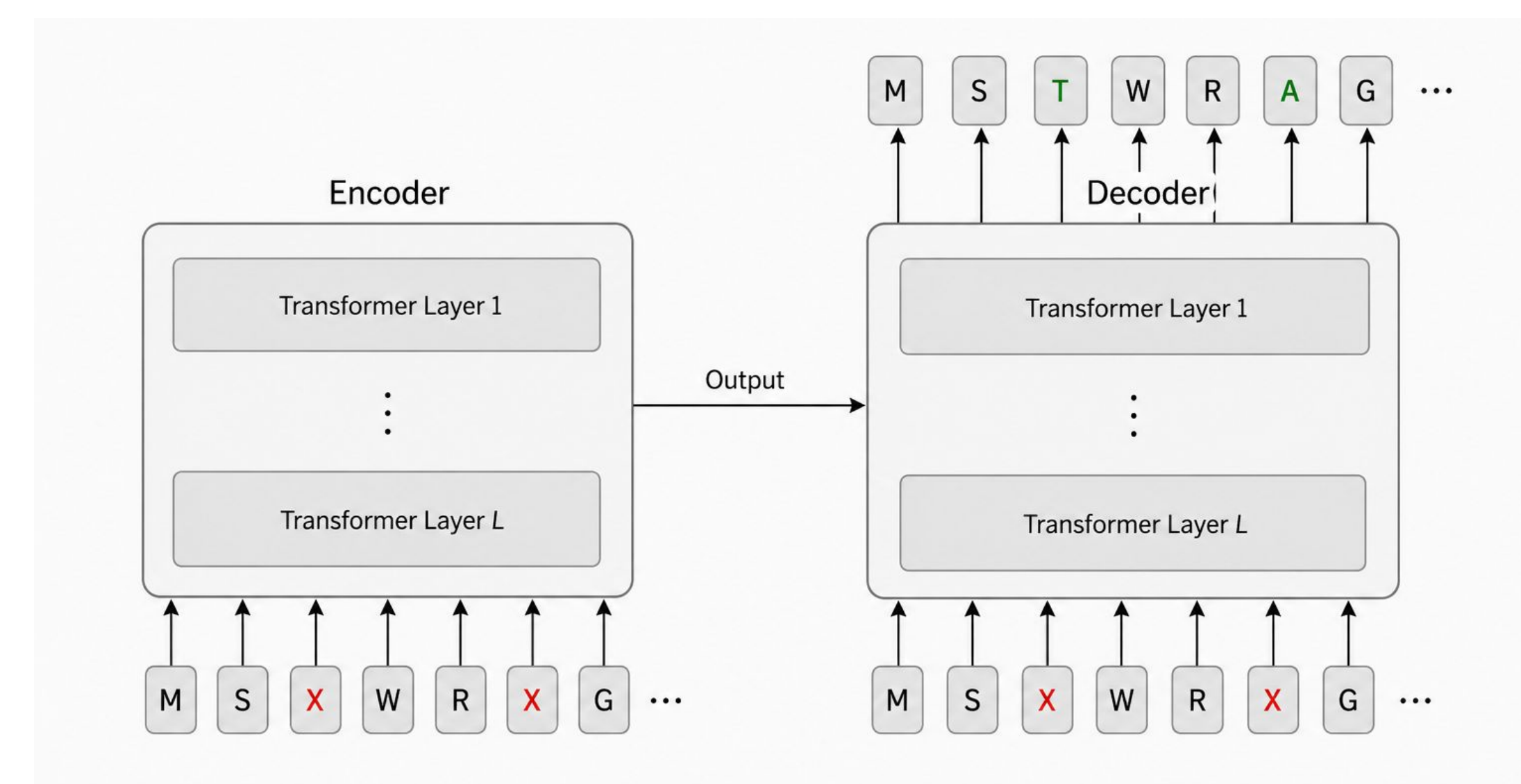
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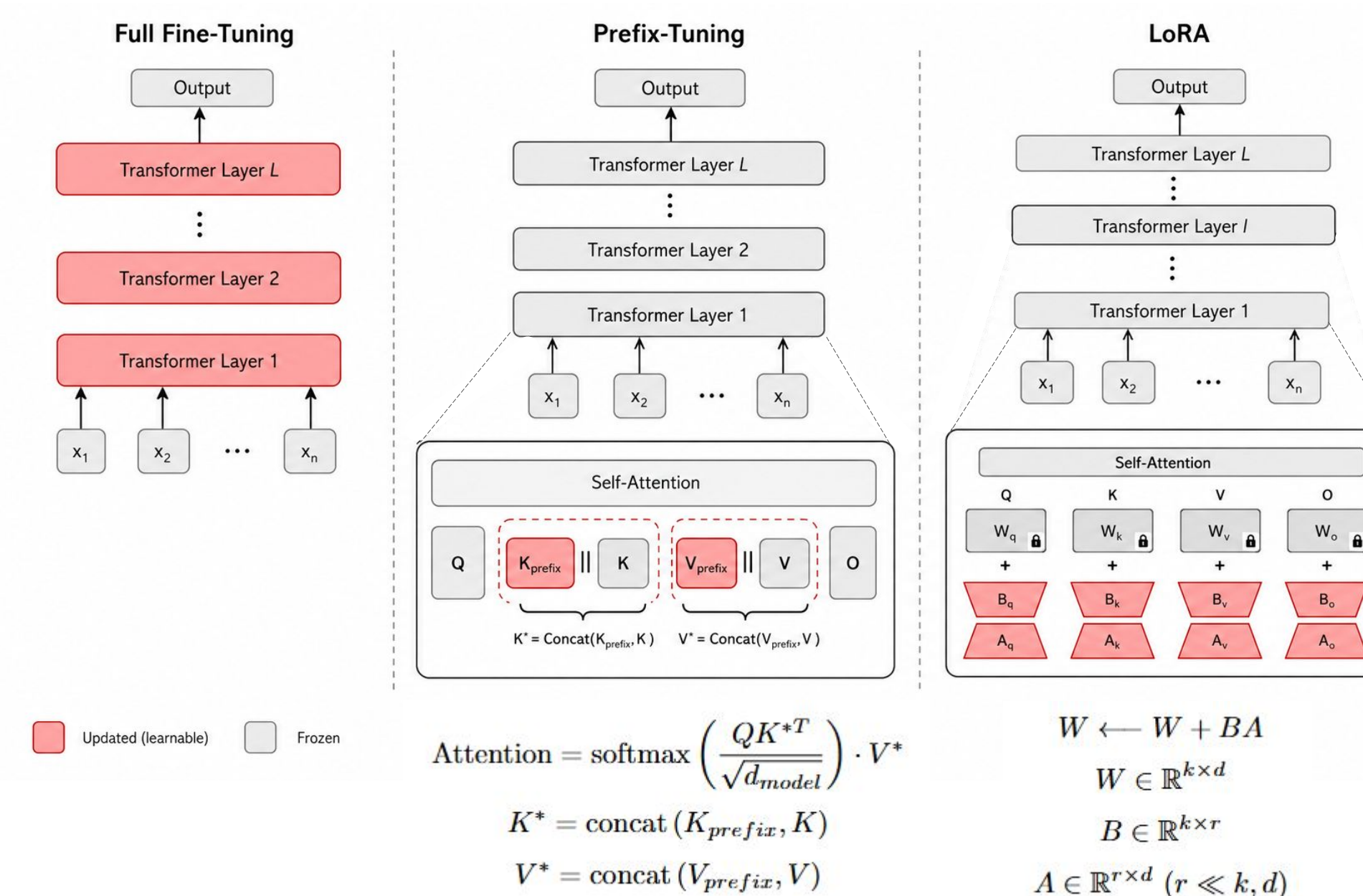
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Architecture



ProtT5 architecture

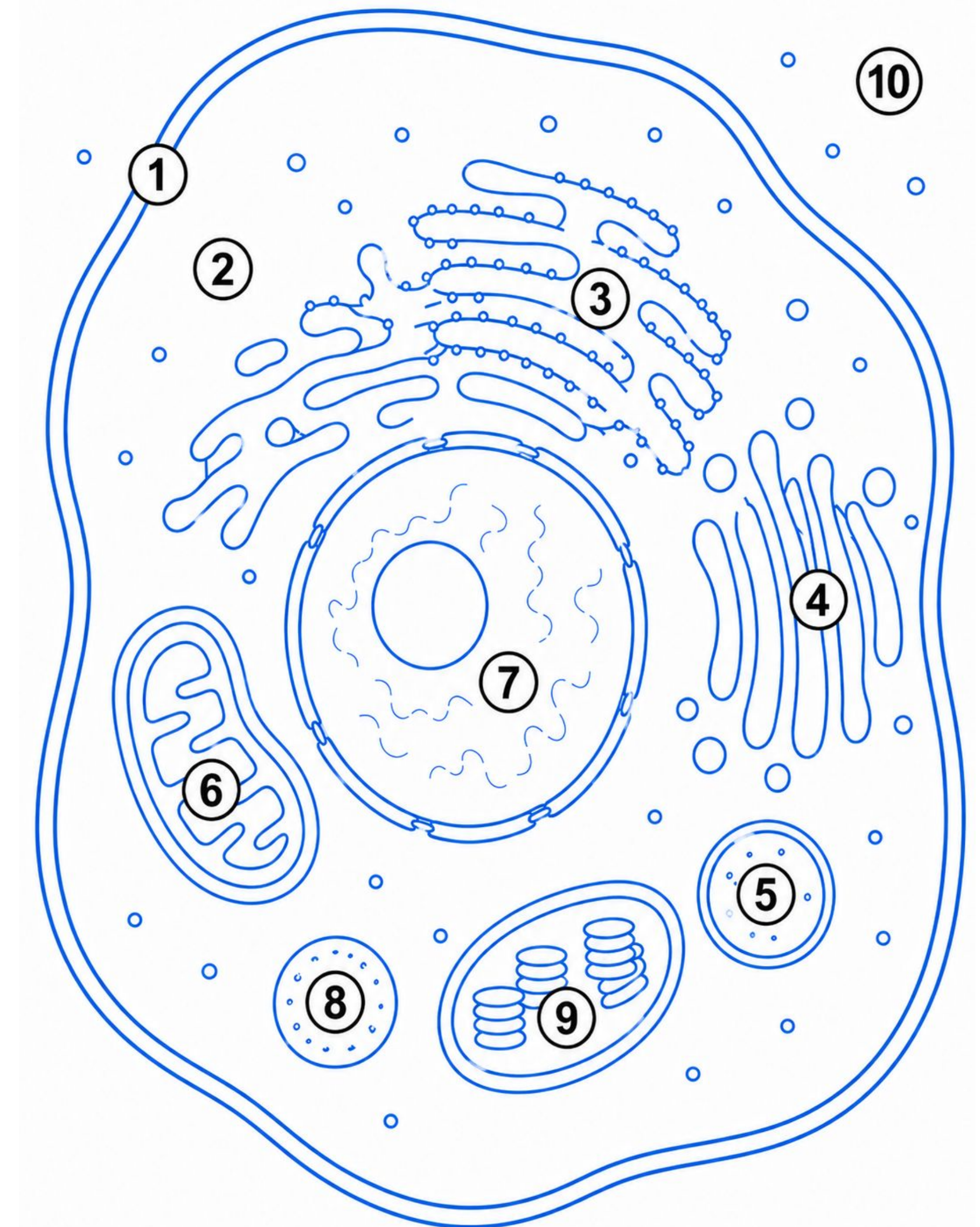


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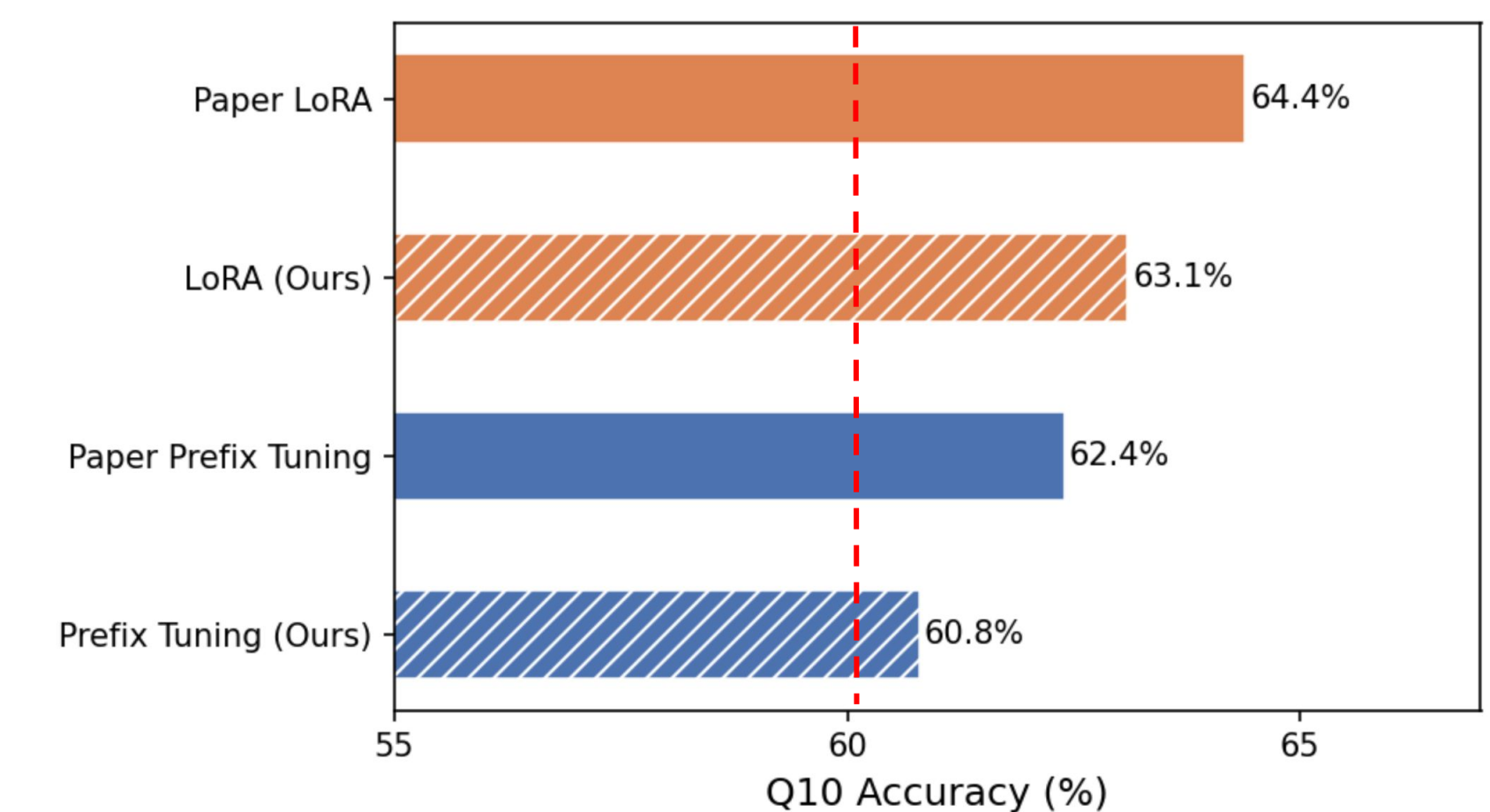
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Trainable parameters for Prefix Tuning and LoRA

Fine-Tuning Task



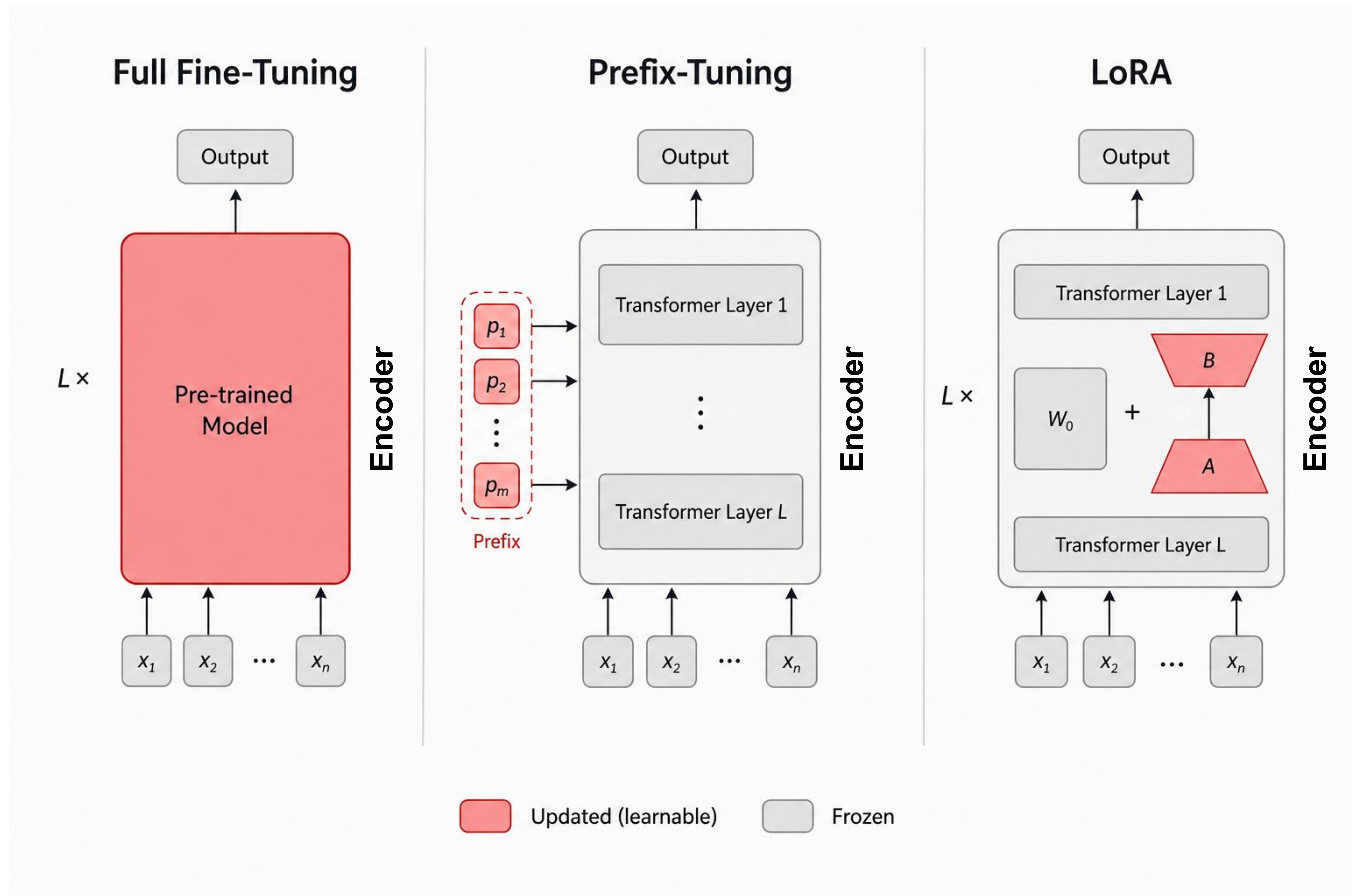
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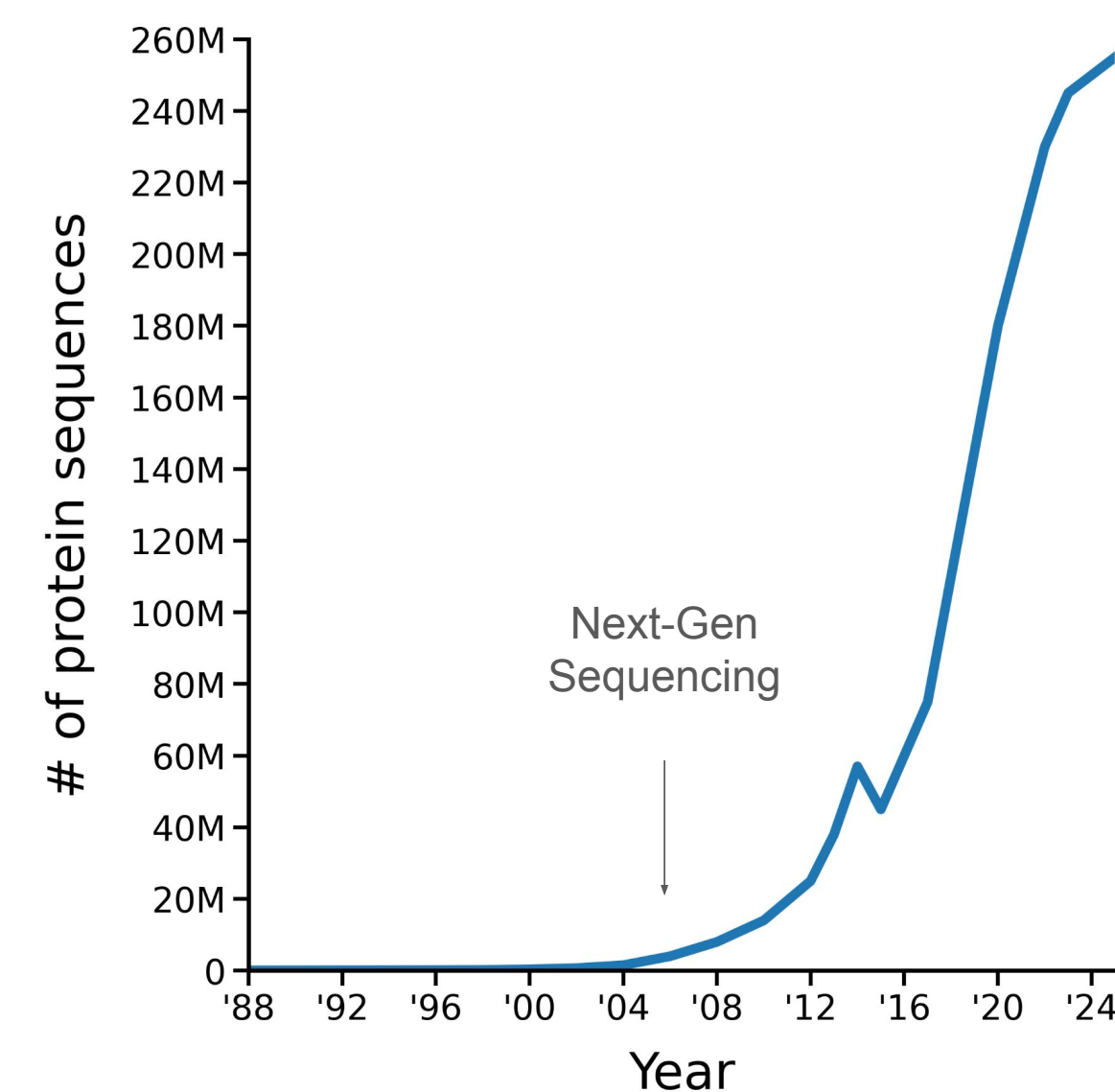
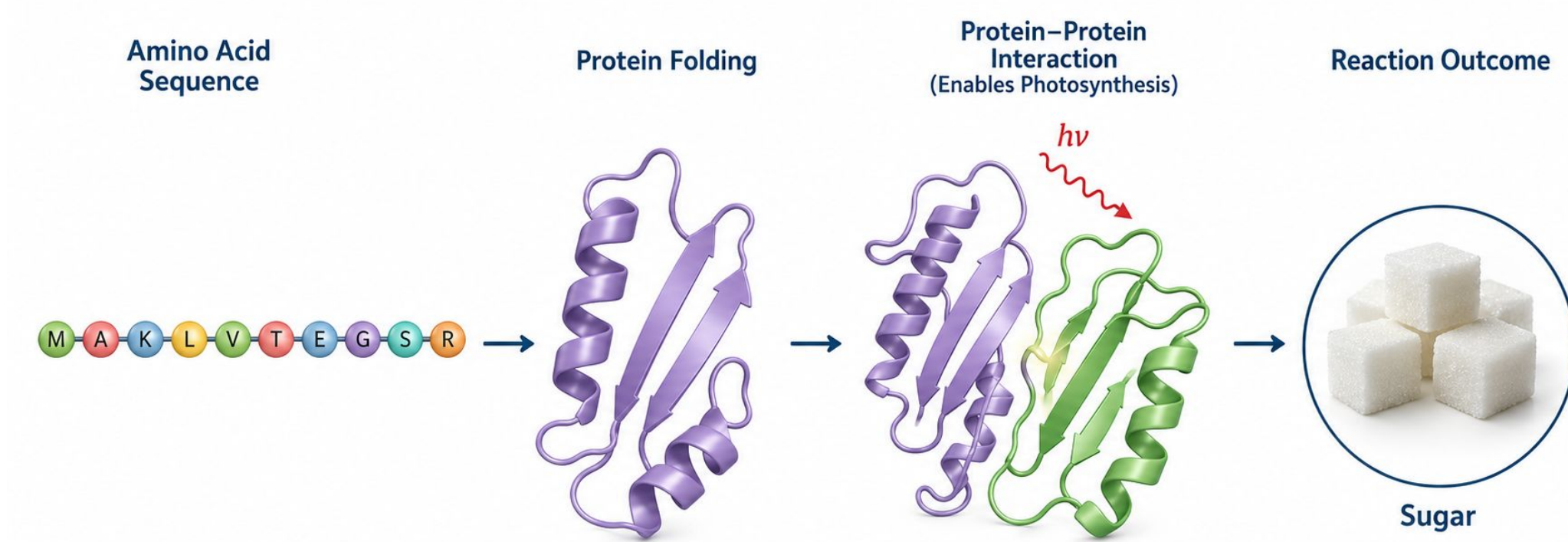
Performance comparison across different schemes, measured in Q10 accuracy. Red vertical line is the baseline.

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Fine-tuning a protein language model to play *pin the protein on the cell*

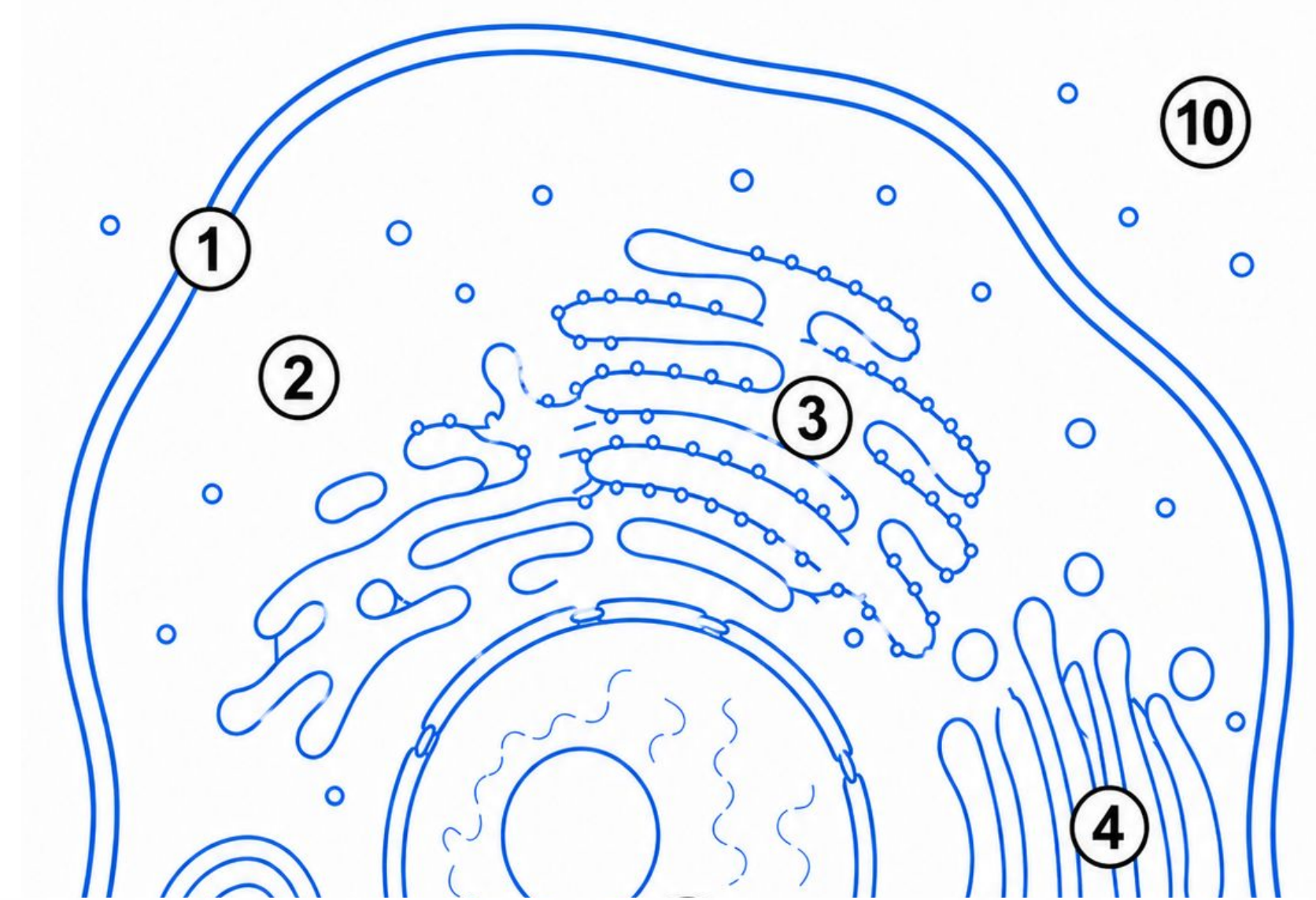
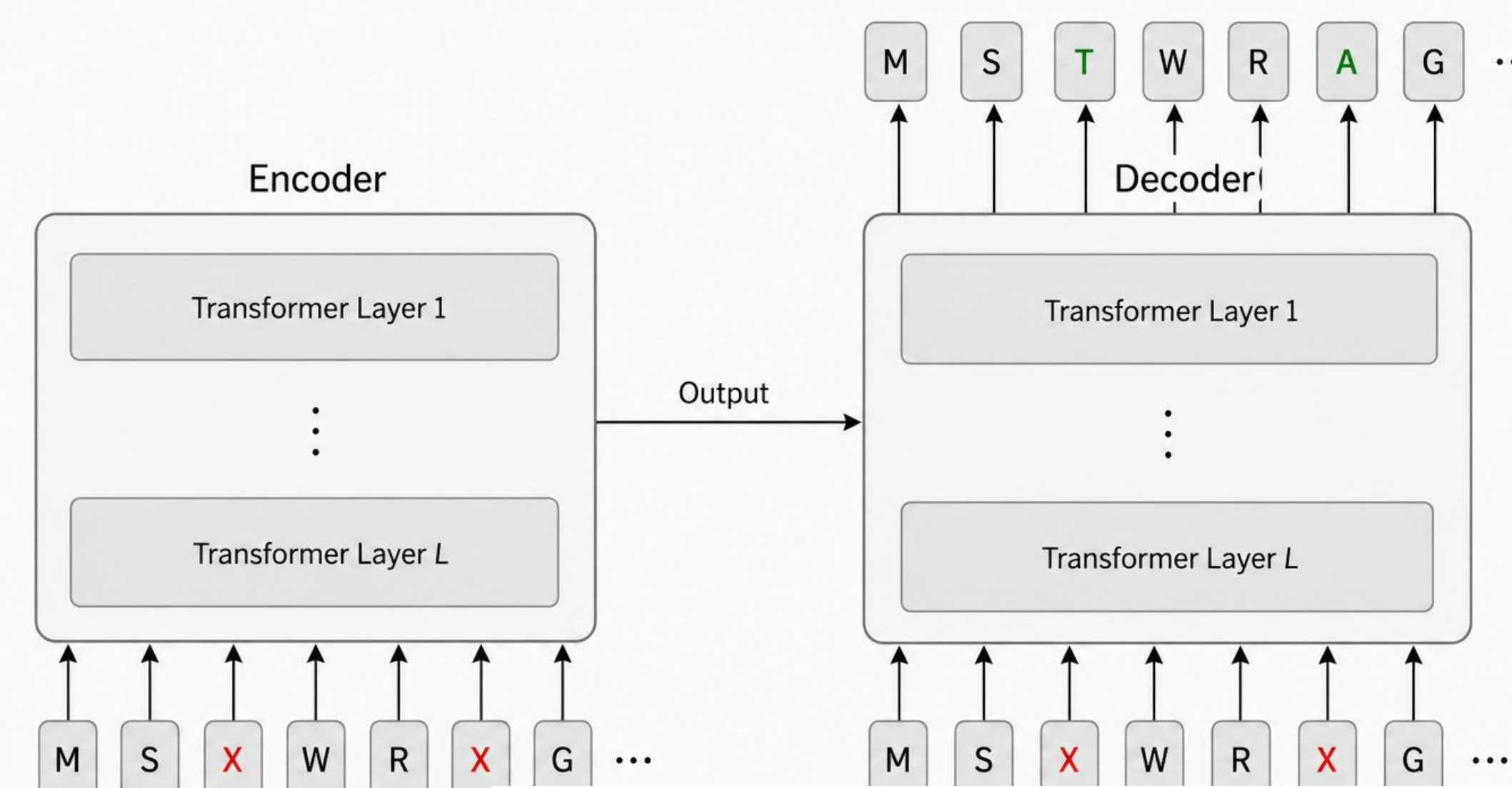


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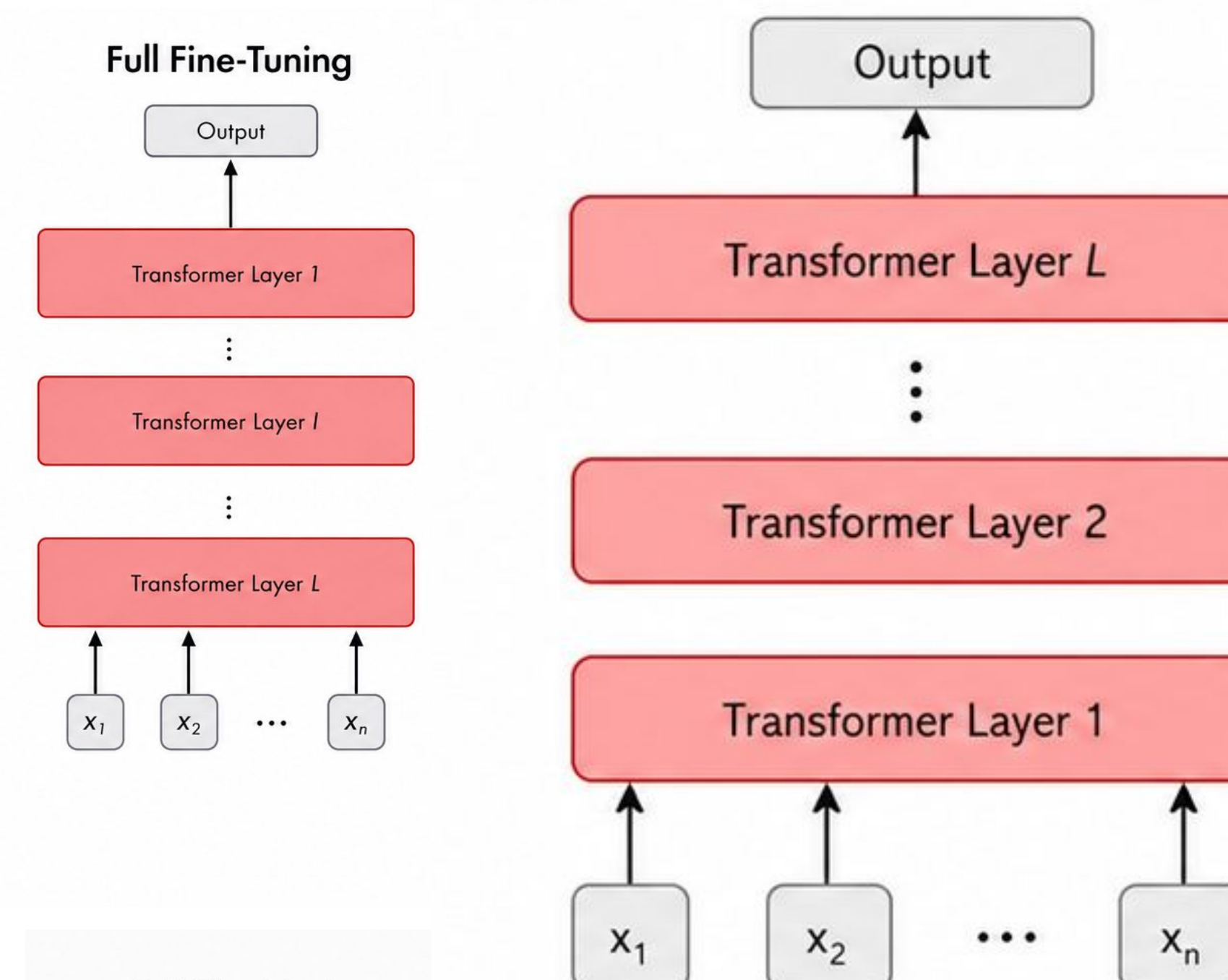


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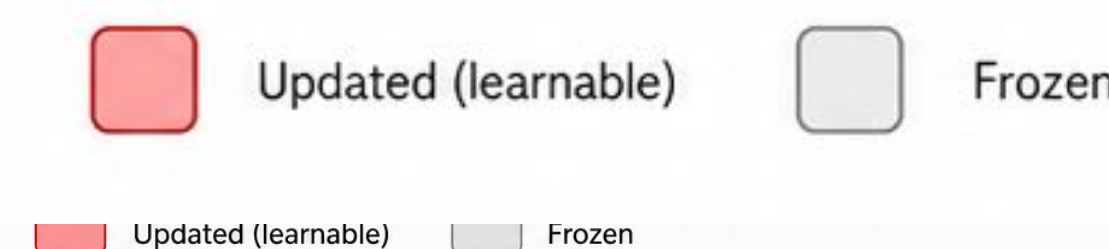
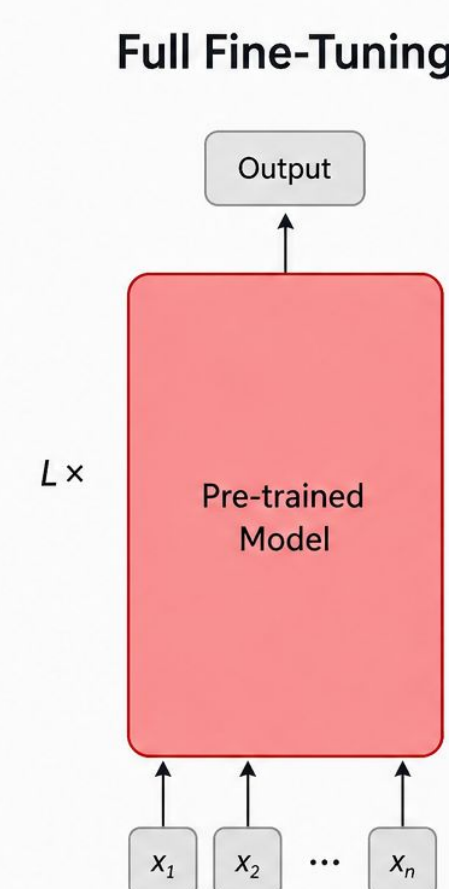
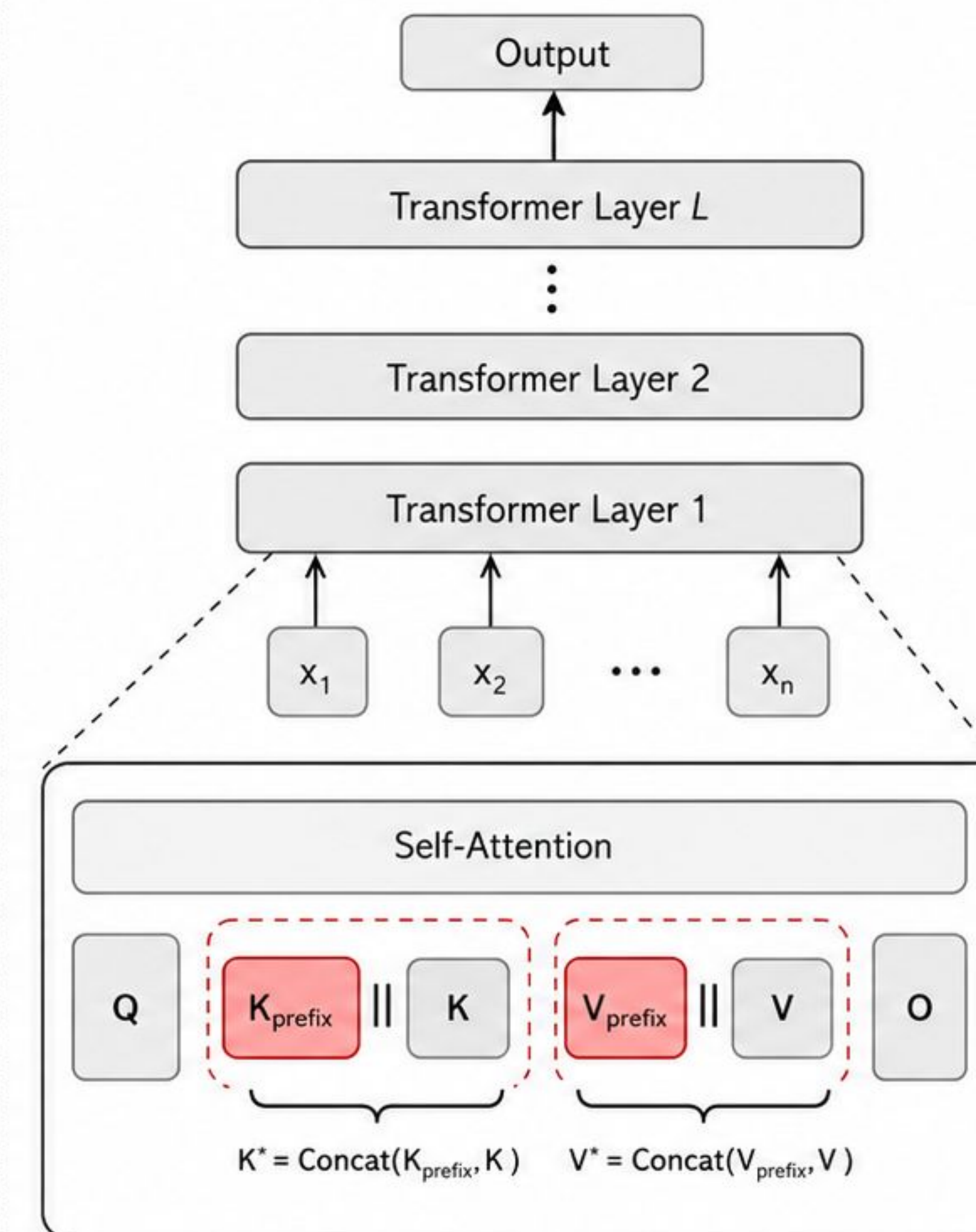
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Full Fine-Tuning

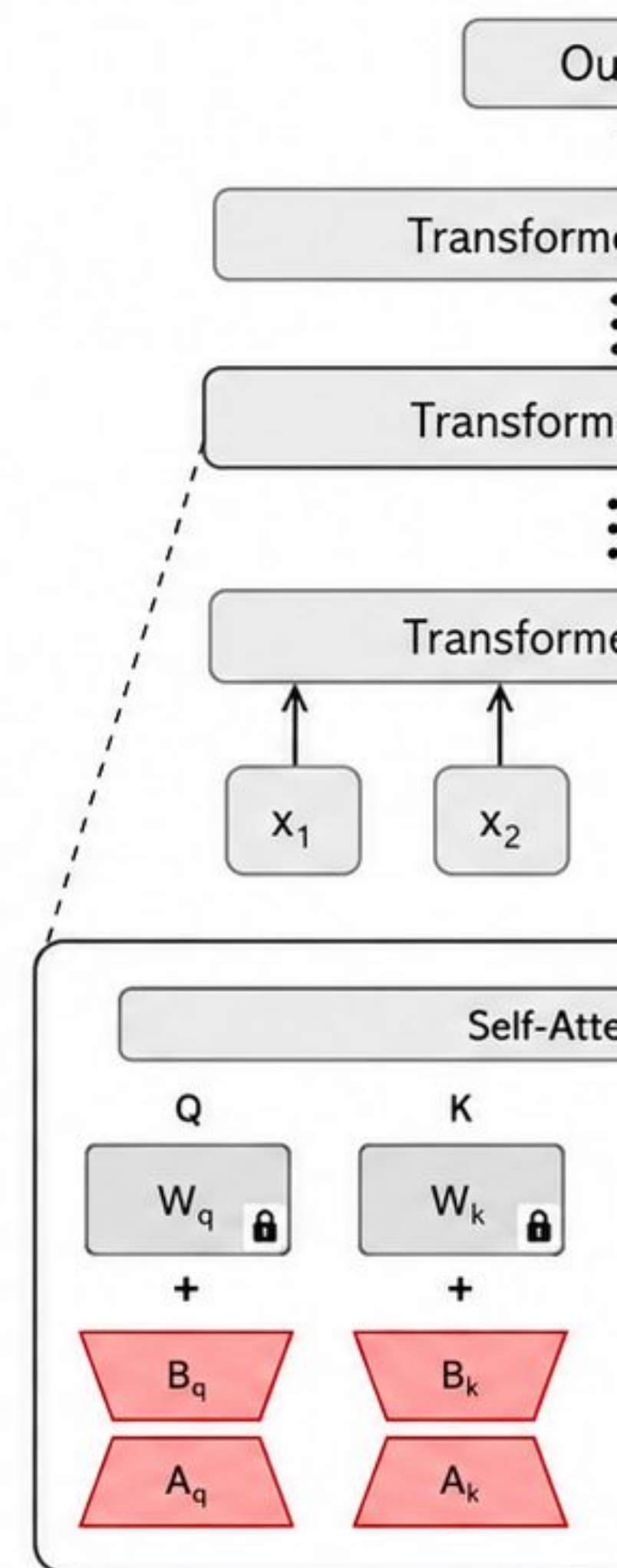


Prefix-Tuning



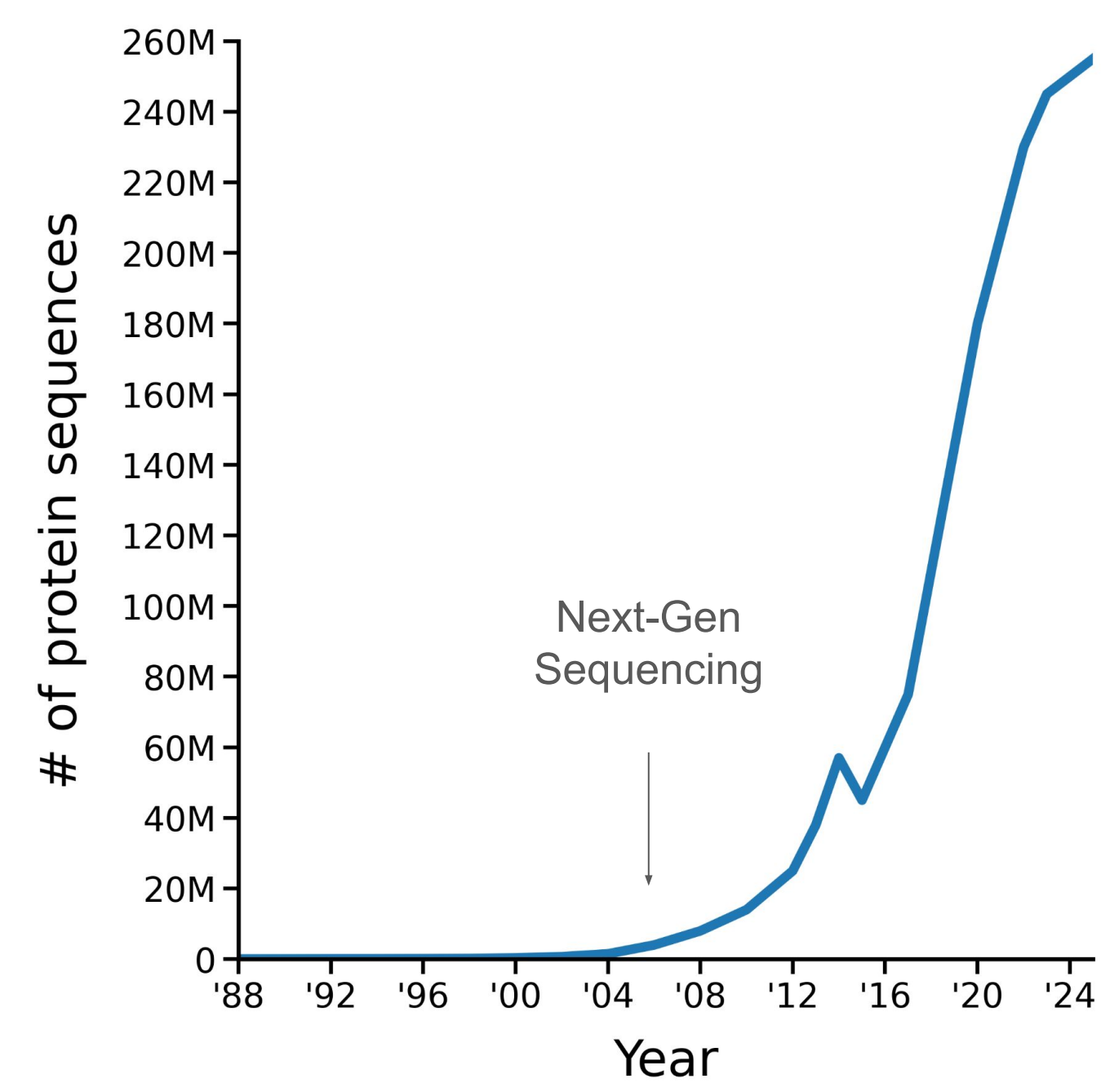
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LoRA

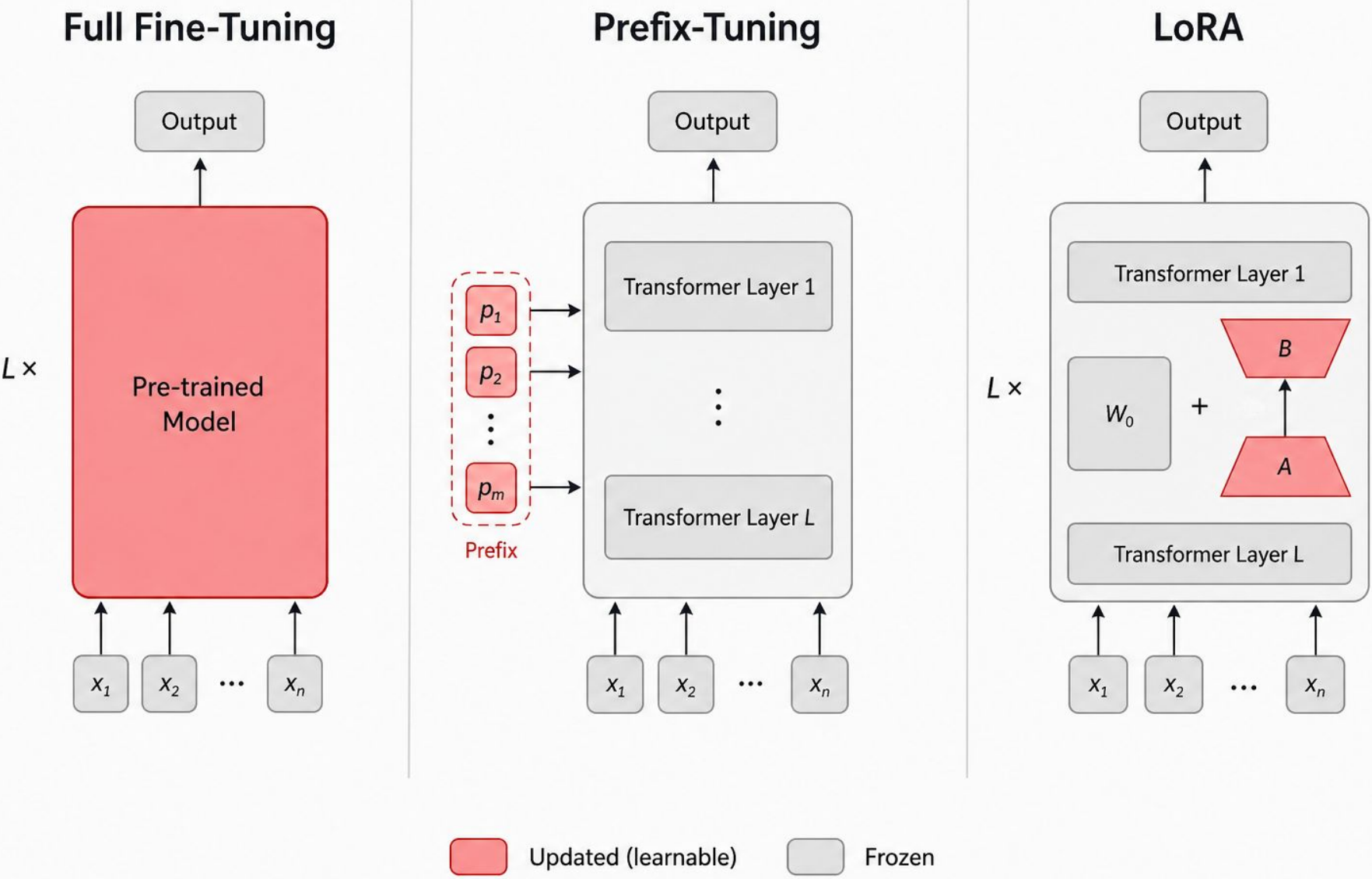
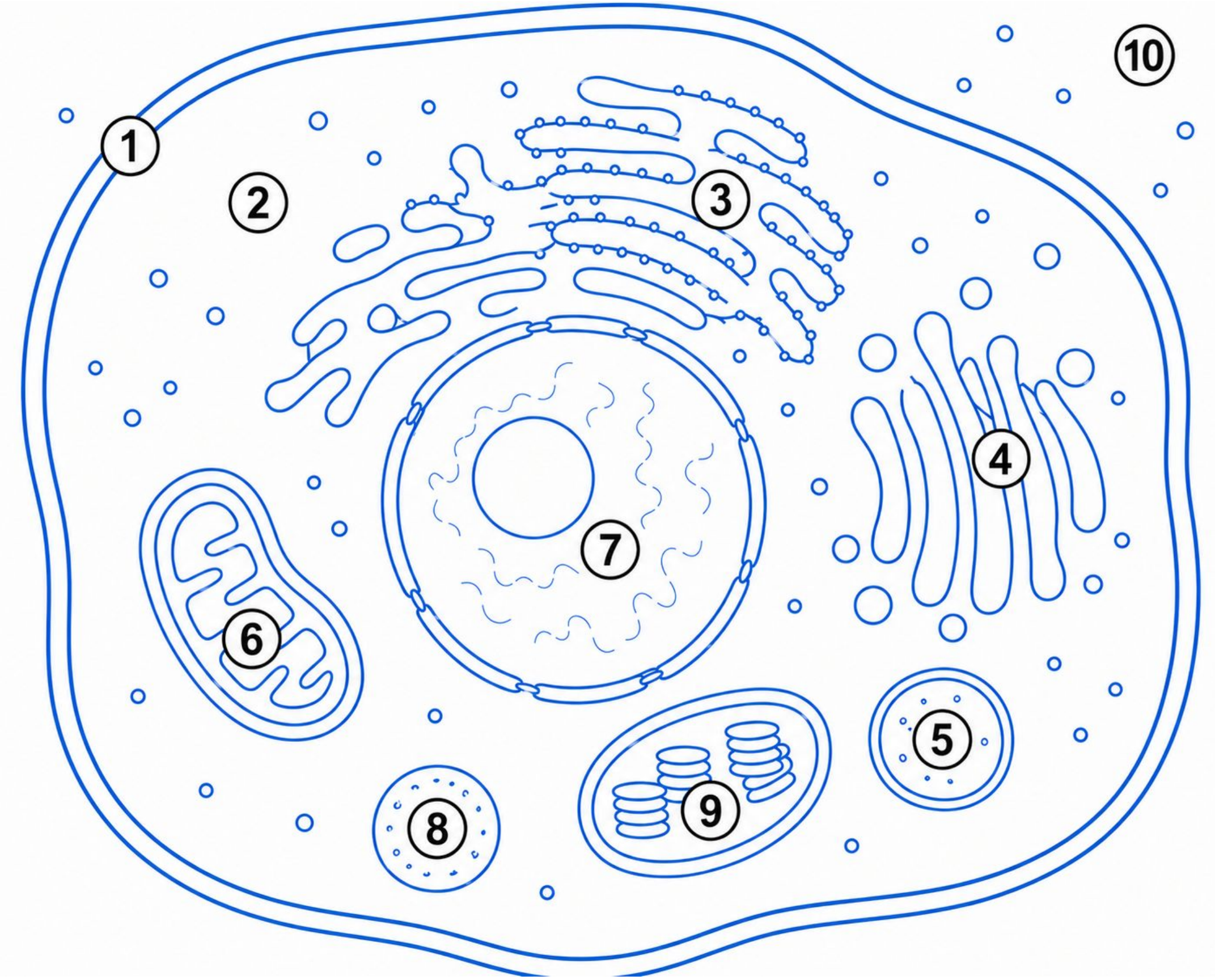


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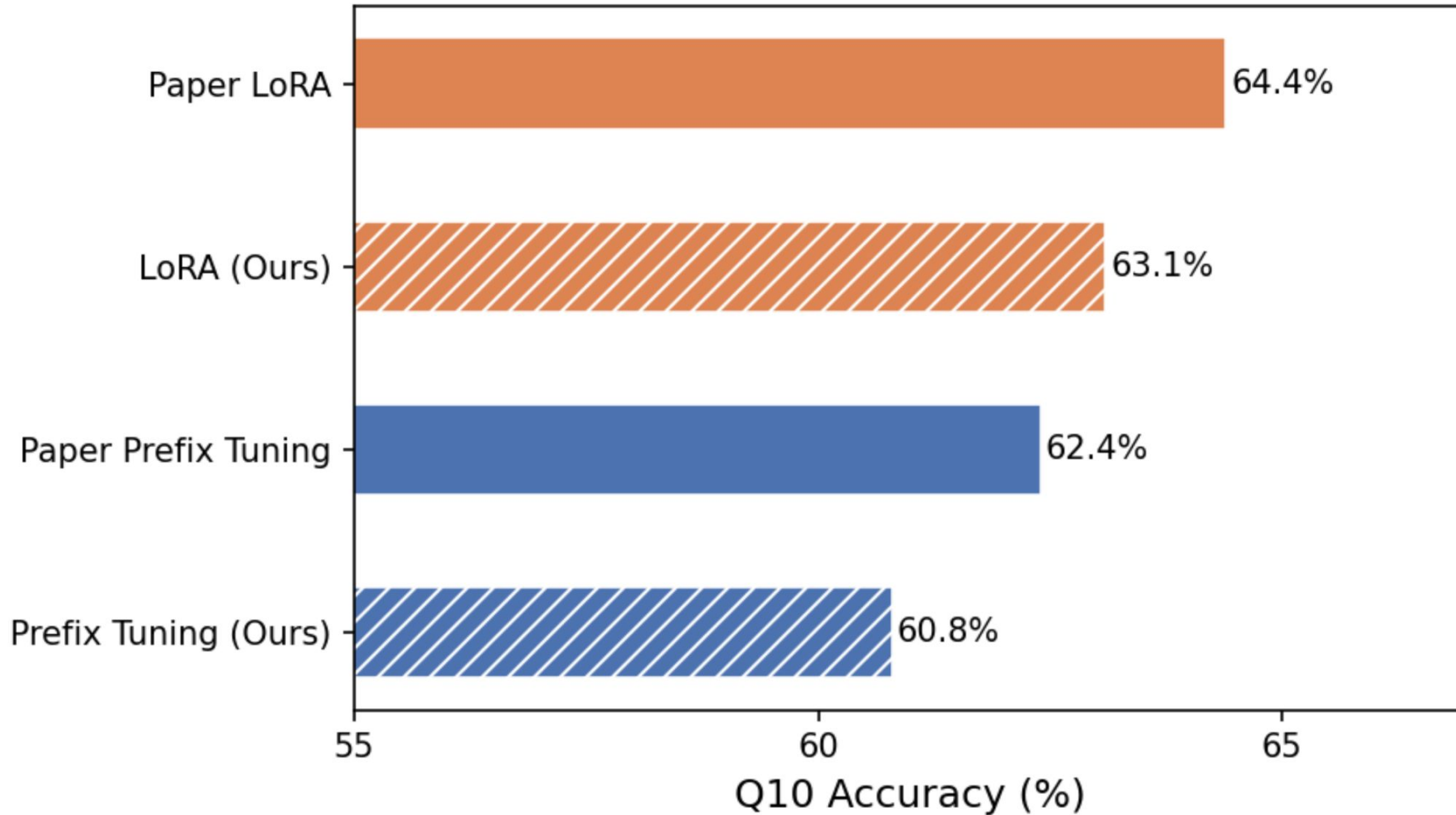
Fine-tuning a protein language model to play *pin the protein on the cell*



Growth of protein sequences in UniProt. Modern protein databases now contain hundreds of millions of sequences, far outpacing experimental labeling and motivating self-supervised protein language models.



1. Cell Membrane
2. Cytoplasm
3. Endoplasmic reticulum
4. Golgi apparatus
5. Lysosome / Vacuole
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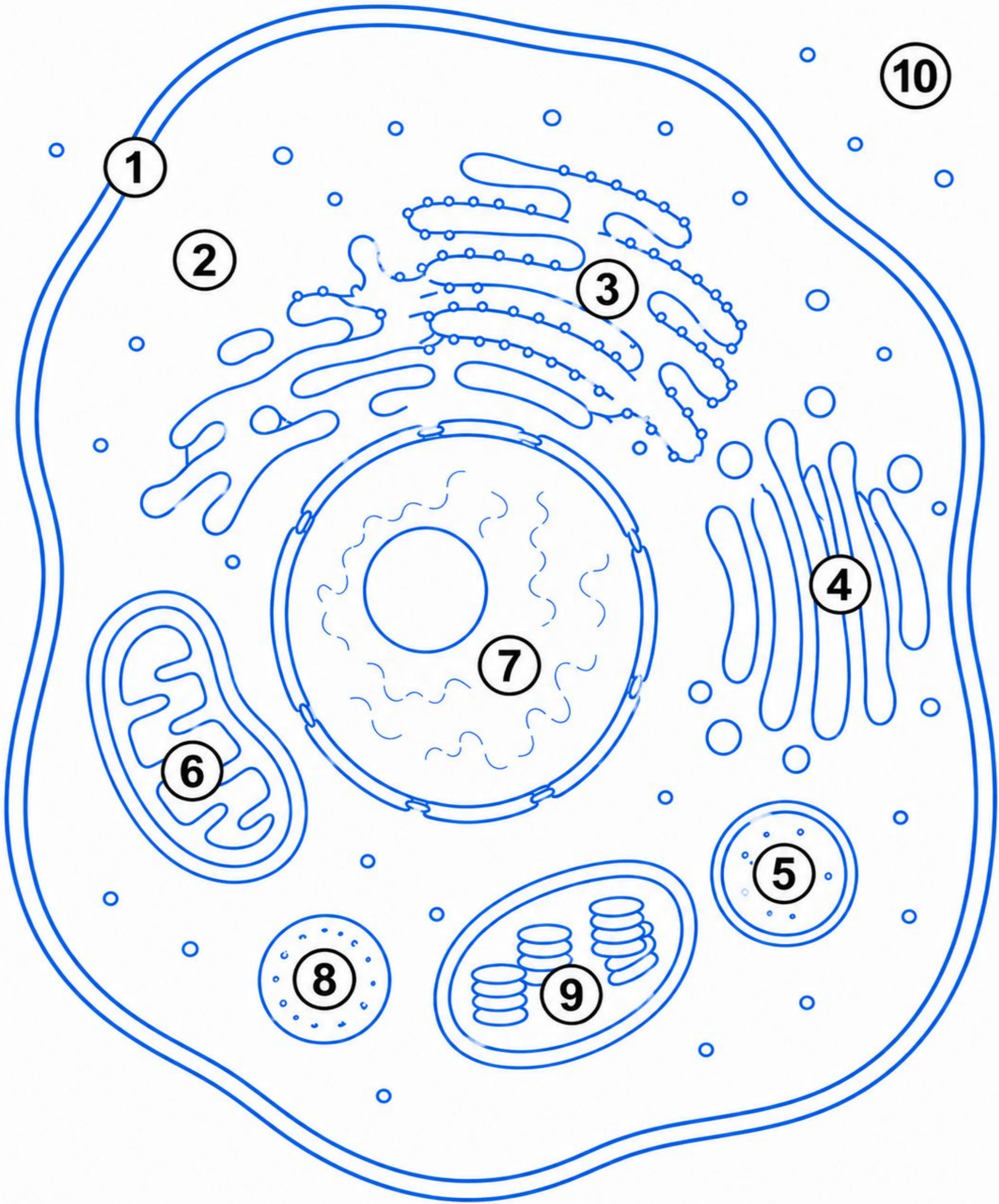
Whats a protein?

Whats a protein sequence?

Protein Language Model

Overview

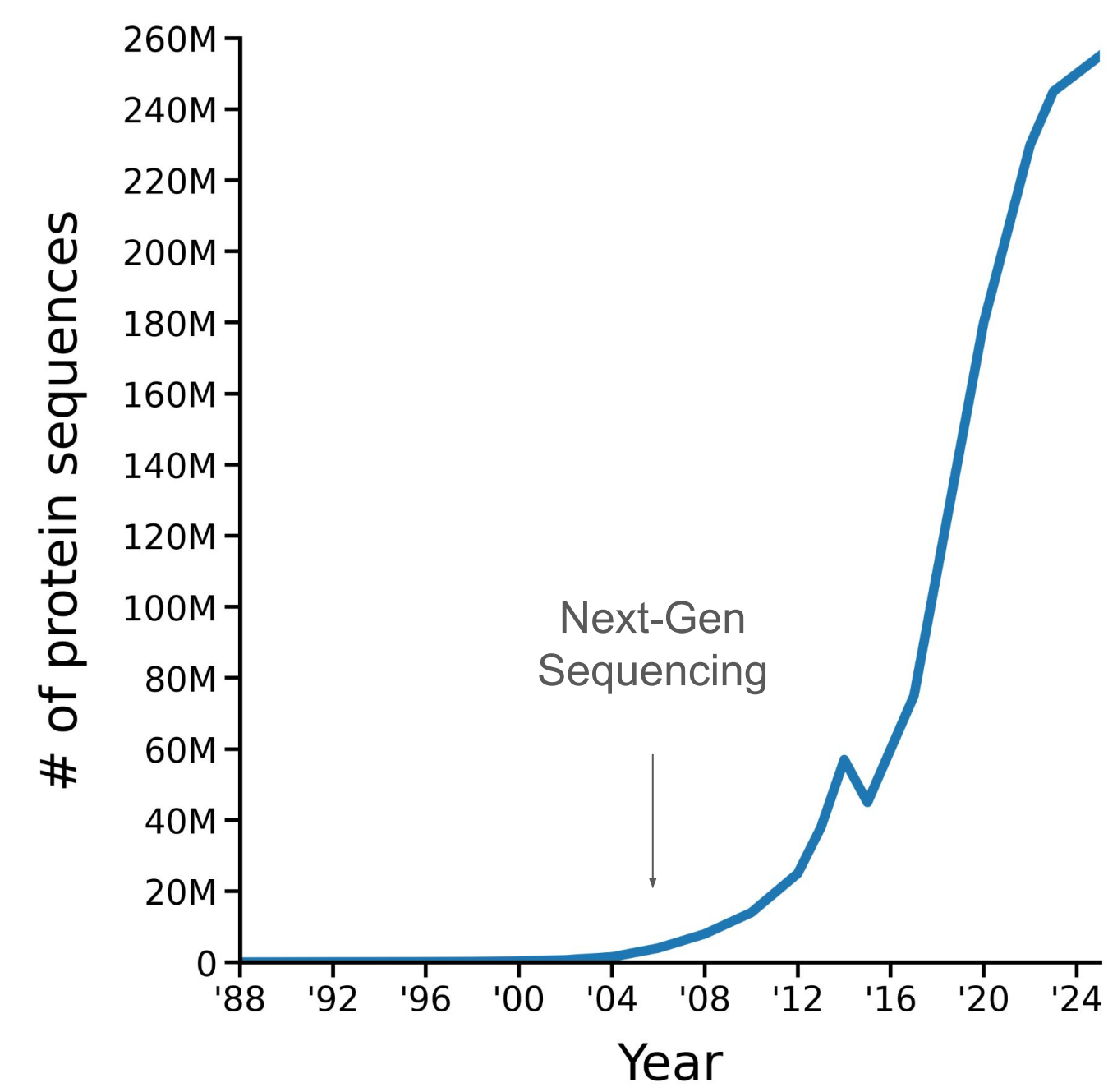
Lora vs prefix tuning vs full finetuning scheme



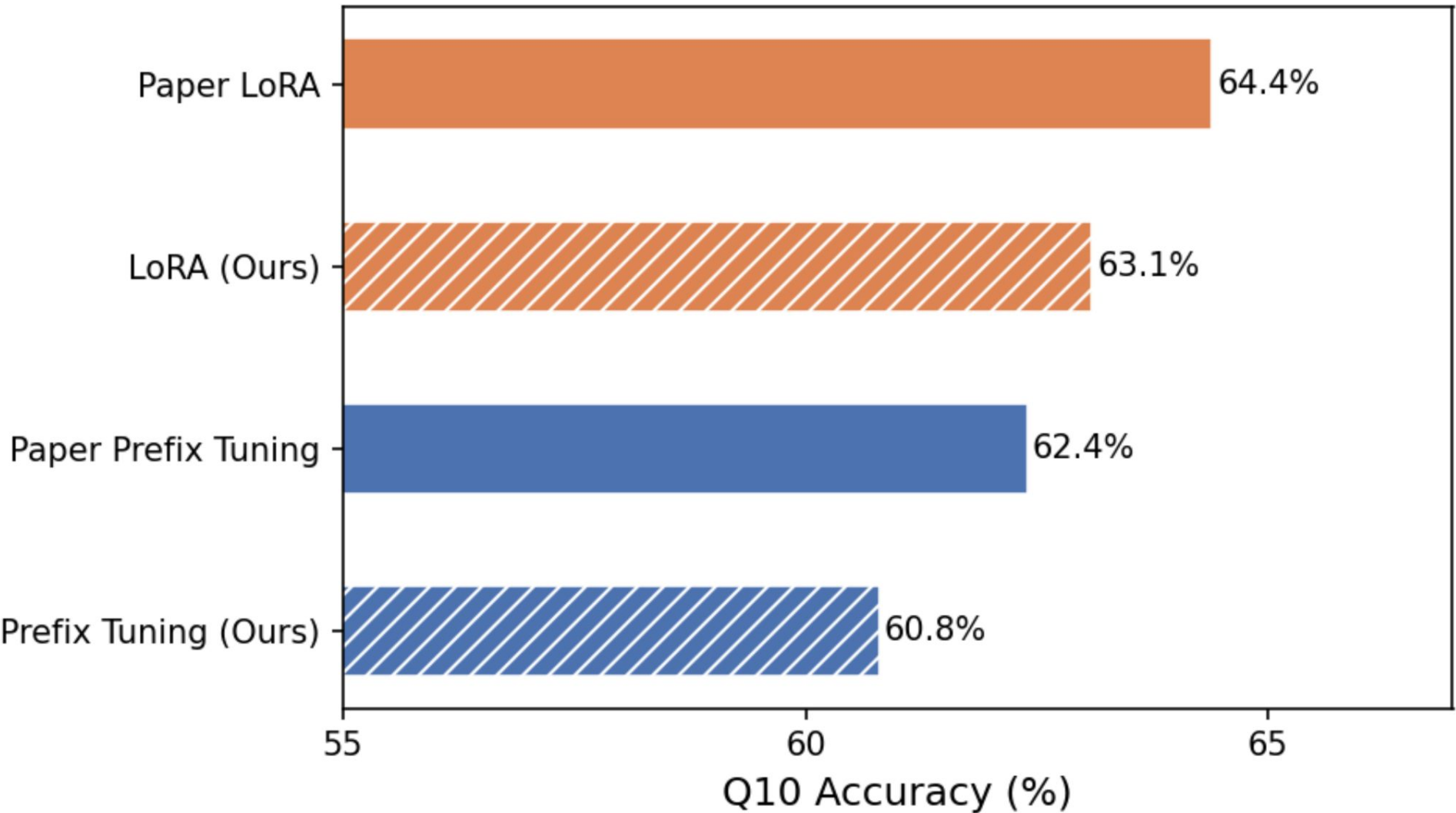
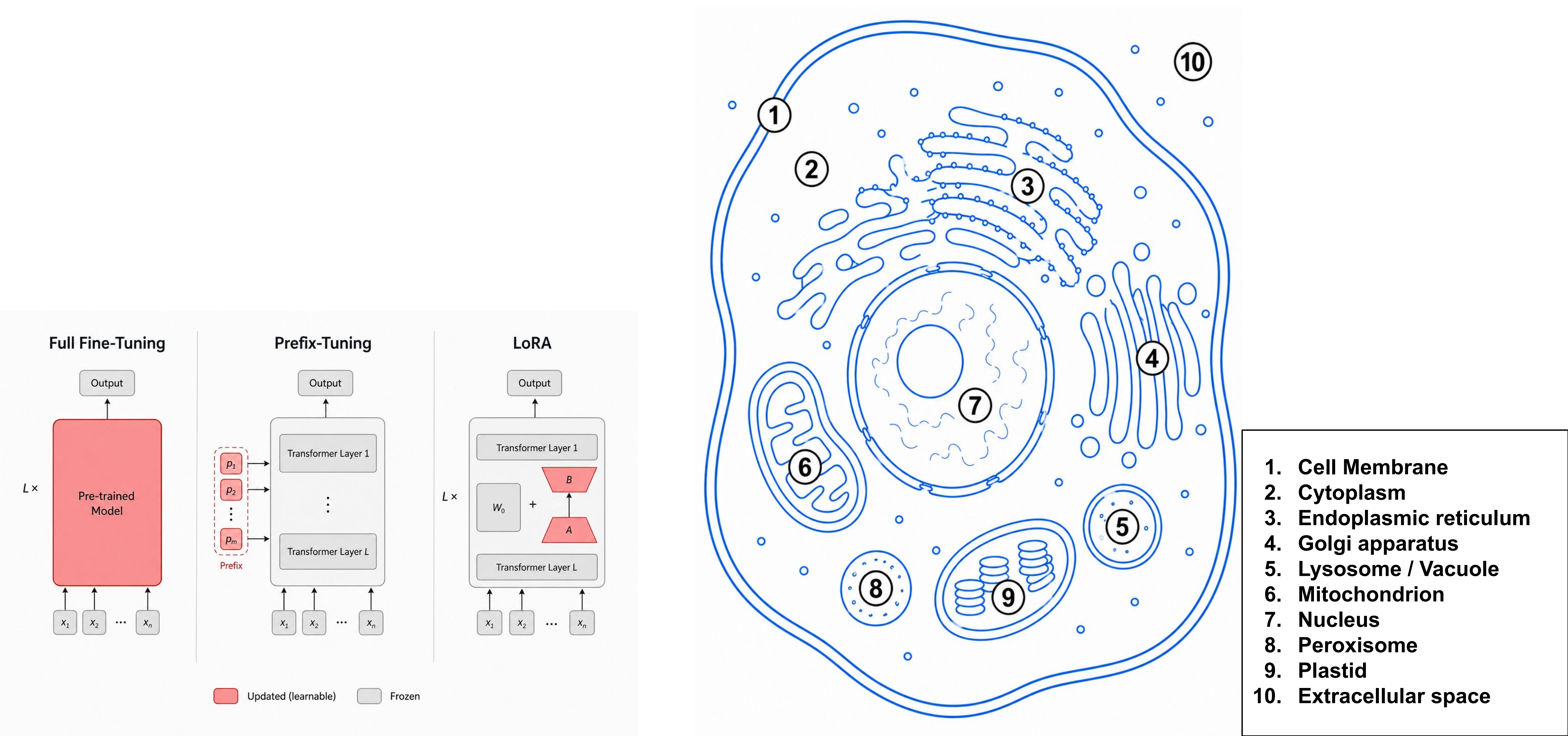
Results

- 1. Cell membrane
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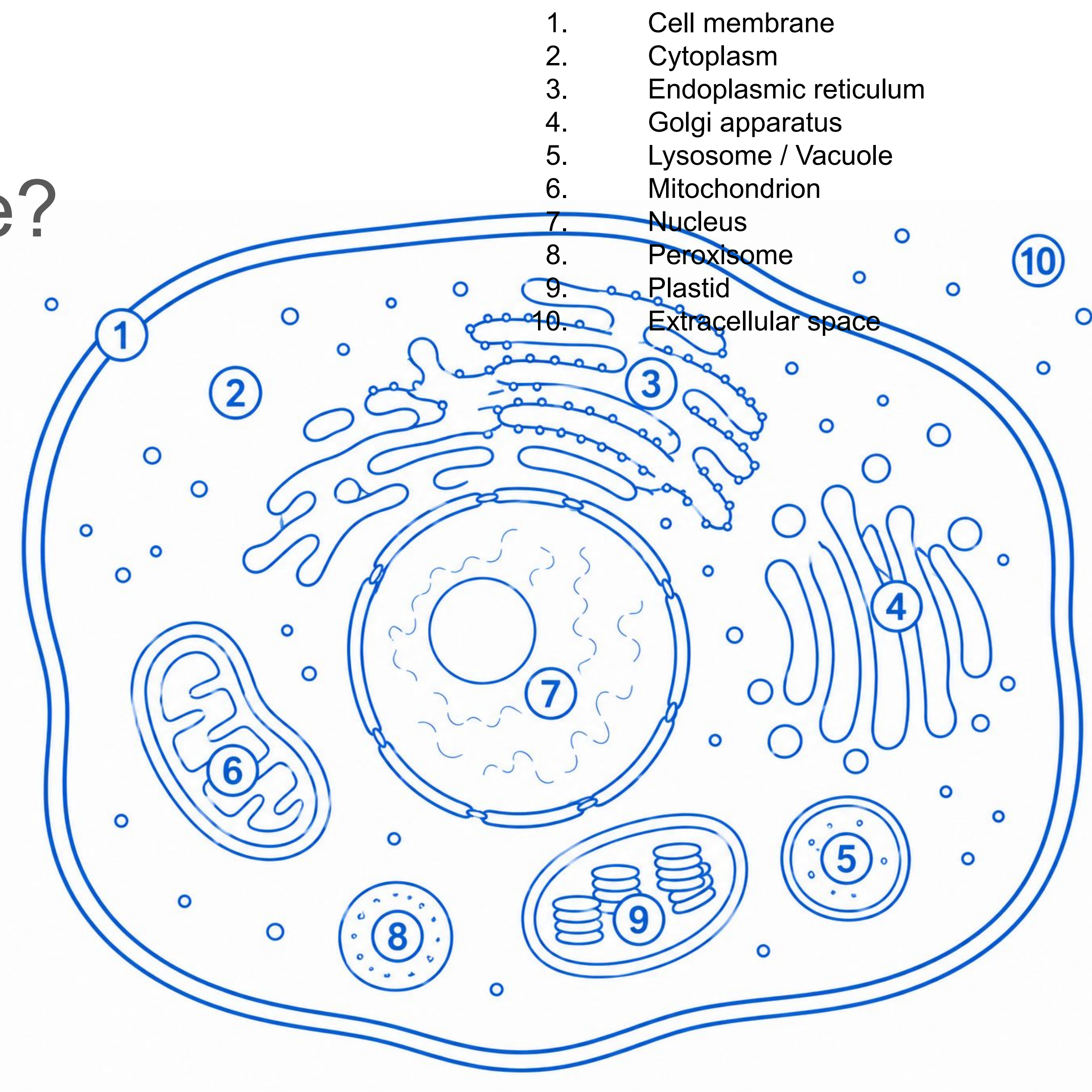
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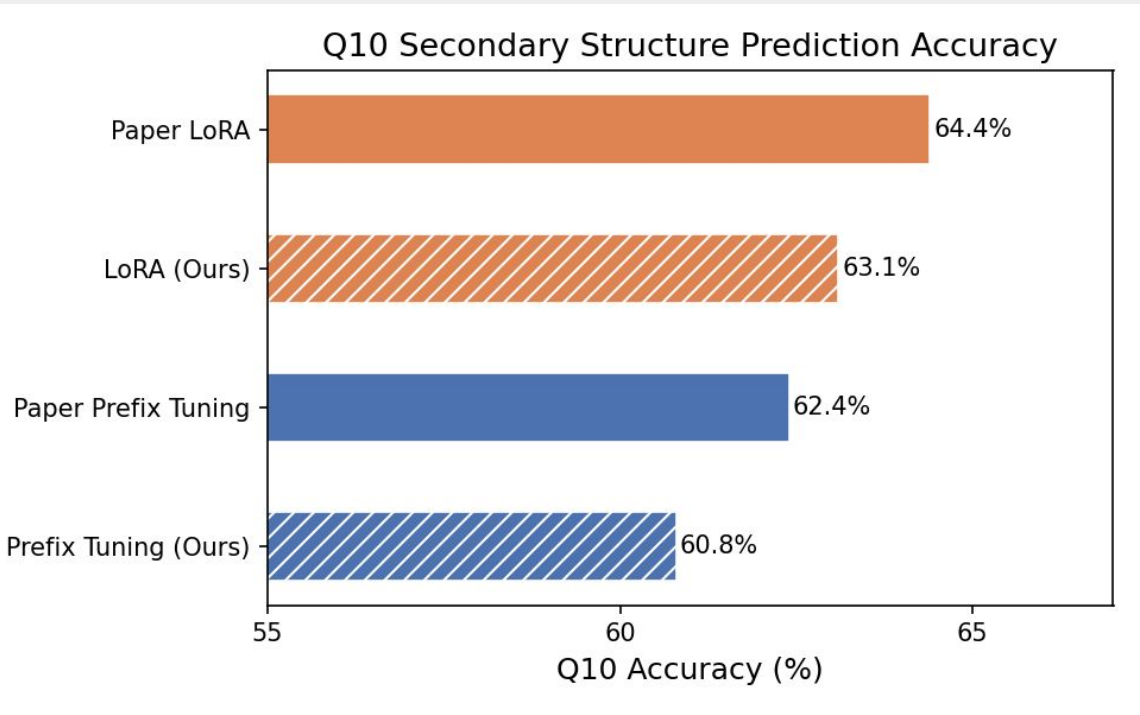
Protein Language Model

Overview

Lora vs prefix tuning vs full finetuning scheme



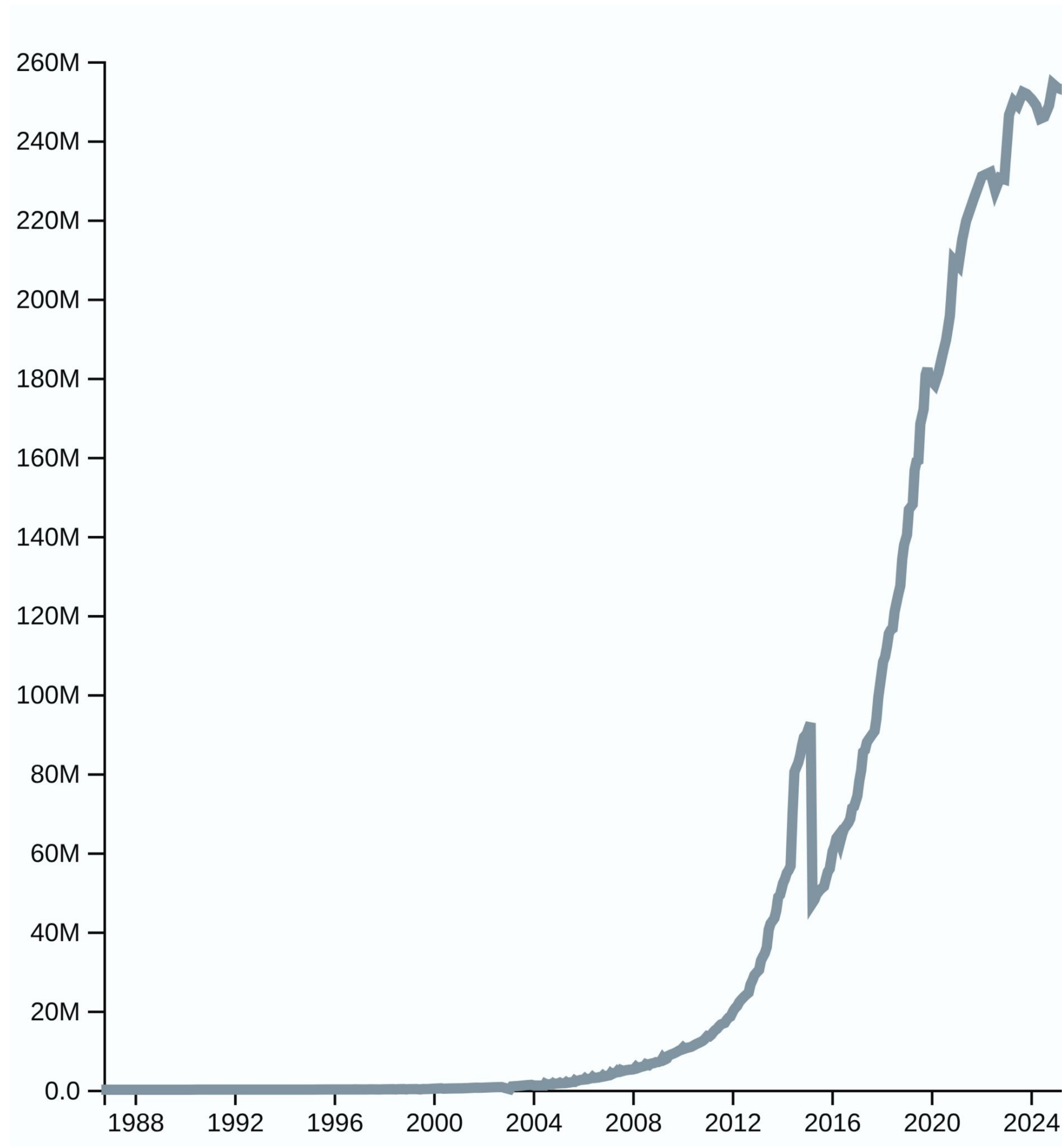
Results



Future Direction

1. Cheap training for screening binders, linkers, pharmaceutical etc..
2. “Semi” method: prefix tuning on ICL (training mixed with no training)

Subcellular localization classification is just pin the *protein* on the *cell*!



Lora vs prefix tuning vs full finetuning scheme

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